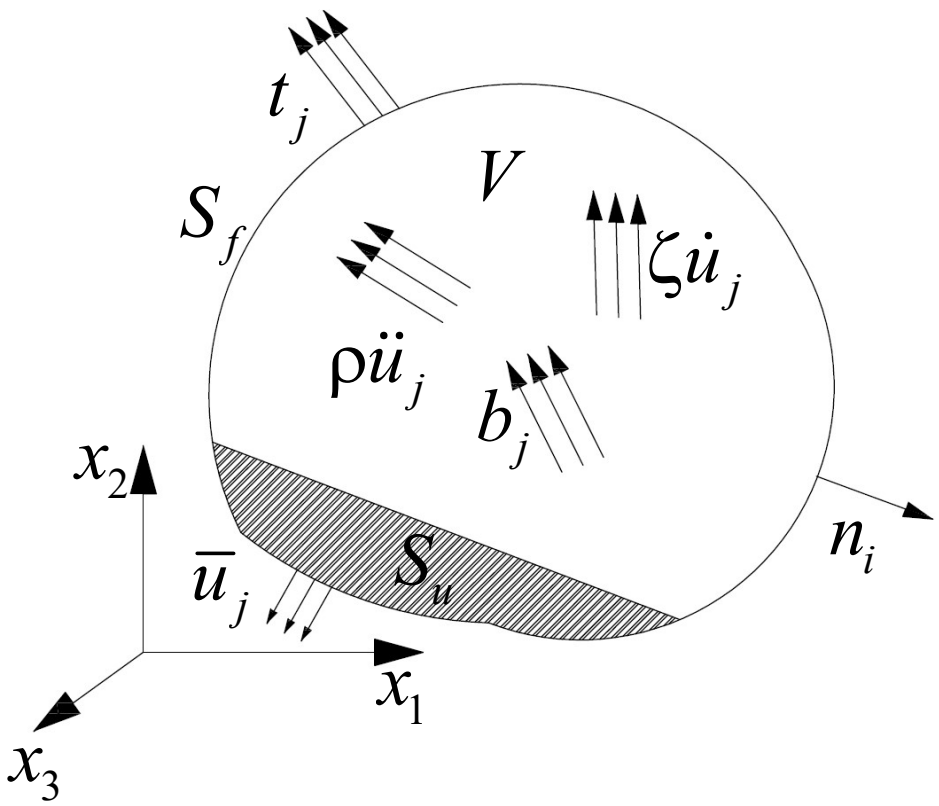
Space discretization of the Elastodynamics problem

Linear elastic Cauchy continuum



$V = volume;$

$S = external\ surface;$

$S_u = constrained\ external\ surface;$

$S_f = loaded\ external\ surface;$

$n_i = unit\ outward\ normal\ vector;$

$b_j = body\ force\ vector;$

$t_j = surface\ traction\ vector;$

$u_j = displacement\ vector;$

$\bar{u}_j = prescribed\ displacement\ vector;$

$\rho\ddot{u}_j = inertia\ force\ vector;$

$\zeta\dot{u}_j = viscous\ force\ vector;$

The Cauchy continuum, in the case of a linear elastic material, and under the assumption of small displacements, is governed by the customary equations of equilibrium, compatibility, and constitutive law, with relevant boundary conditions.

In Dynamics, only equilibrium must be rewritten, in order to account for the inertia forces and, possibly, for damping terms. Initial conditions must be added, too.

Dynamic equilibrium (with viscous damping)

$$\frac{\partial \sigma_{ij}\left(x_k,t\right)}{\partial x_i}+b_j\left(x_k,t\right)=\rho\left(x_k\right)\ddot{u}_j\left(x_k,t\right)+\zeta\left(x_k\right)\dot{u}_j\left(x_k,t\right)\ \forall t \text{ in } V;$$

$$\sigma_{ij}\left(x_k,t\right)n_i\left(x_k\right)=t_j\left(x_k,t\right)\ \forall t \text{ in } S_f$$

where ρ is the material density and ζ the viscous damping coefficient.

Compatibility and boundary conditions

$$\varepsilon_{ij}\left(x_k,t\right)=\frac{1}{2}\left(\frac{\partial u_i\left(x_k,t\right)}{\partial x_j}+\frac{\partial u_j\left(x_k,t\right)}{\partial x_i}\right)\ \forall t \text{ in } V;$$

$$u_j\left(x_k,t\right)=\bar{u}_j\left(x_k,t\right)\forall t \text{ in } S_u;$$

Initial conditions

$$u_j\left(x_k,0\right)=u_j^{\,0}\left(x_k\right)\ \text{for } t=0 \text{ in V;}$$

$$\dot{u}_j\left(x_k,0\right)=\dot{u}_j^{\,0}\left(x_k\right)\ \text{for } t=0 \text{ in V;}$$

Direct linear elastic constitutive law

$$\sigma_{ij}\left(x_\ell,t\right)=D_{ijhk}\left(x_\ell\right)\varepsilon_{hk}\left(x_\ell,t\right)\quad \forall t \text{ in } V$$

Inverse linear elastic constitutive law

$$\varepsilon_{ij}\left(x_\ell,t\right)=C_{ijhk}\left(x_\ell\right)\sigma_{hk}\left(x_\ell,t\right)\quad \forall t \text{ in } V$$

This system of linear partial differential equations and linear algebraic ones can be formulated in several alternative ways, eliminating some unknowns as functions of others.

One such technique was proposed by Claude-Louis **Navier** in 1821. In it, strains and stresses are eliminated from the problem using the compatibility and constitutive law equations, to express them as function of the displacements only. The expression of stresses as functions of displacements is finally replaced into the equilibrium equations – the only remaining – obtaining a formulation in the displacements only, called **Navier equations**:

$$\frac{\partial^2 u_i}{\partial x_1{}^2}+\frac{\partial^2 u_i}{\partial x_2{}^2}+\frac{\partial^2 u_i}{\partial x_3{}^2}+\frac{1}{1-2\nu}\frac{\partial^2 u_j}{\partial x_i \partial x_j}+\frac{b_i}{G}=\frac{\rho}{G}\frac{\partial^2 u_i}{\partial t^2}+\frac{\zeta}{G}\frac{\partial u_i}{\partial t} \quad \text{in } V$$

In Elastodynamics it is also possible to write a **Principle of Virtual Work (PVW)** as follows:

$$\int_V b_j u_j^V dV + \int_{s_f} t_j u_j^V dV + \int_{S_u} \sigma_{ij} n_i \bar{u}_j^V dS =$$

$$\int_V \sigma_{ij} \varepsilon_{ij}^V dV + \int_V \rho \ddot{u}_j u_j^V dV + \int_V \zeta \dot{u}_j u_j^V dV$$

In this, the quantities with apex ^{V} are virtual quantities, i.e., infinitesimal and compatible. They are not related to the actual loads.

This scalar equation can be seen as an identity if both equilibrium and compatibility are satisfied a priori;
but it can also be used as an alternative version of the equilibrium equations if only compatibility is prescribed a priori; or, vice-versa, as an alternative compatibility equation if only equilibrium is enforced a priori.

Since we are seldom capable to solve in closed form this type of problem, we must often resort to numerical (approximate) techniques, well studied by people in Numerical Analysis.
All these techniques take the initially continuous problem, both in space and in time, and make it discrete, both in space and in time.
In the case of the Elastodynamics equations, the discretization is performed in two separate, sequential steps. First the problem is discretized in space; once it has been transformed into the familiar Dynamics MDOF one:

$$\mathbf{M}\ddot{\delta}(t)+\mathbf{C}\dot{\delta}(t)+\mathbf{K}\delta(t)=\mathbf{F}(t)$$

then some technique, exact or approximate, is further adopted to compute vector $\delta(t)$.

The techniques adopted to compute the vector of the degrees of freedom d(t) can be of several types. Among them:

- the modal superposition technique, which exploits the method we have seen with reference to the analysis of MDOF systems;
- numerical direct time integration methods.

Therefore, here we need to start giving a look at the most commonly adopted technique for the space discretization of the Elastic (and also of the Elastodynamics) Problem: the so-called Finite Element method.
Once we have learned how to make the problem discrete in space, we will study some numerical techniques to solve it approximately also in time.

The numerical methods for the solution of partial differential equations are fundamentally different if the starting differential problems are

- with initial conditions, or
- with boundary conditions.

The space discretization of the Elastic (or Elastodynamics) Problem, of present interest to us, concerns a differential problem with boundary conditions.
The starting physical problem could be formulated and solved, in principle, both
(a) in terms of equations and
(b) in terms of either a variational formulation, if it exists, or a weak form, such as the PVW.
There exist techniques for the numerical solution of both types (a) and (b) of formulations.

A “standard” method of space discretization that operates directly on the differential equations:

the Finite Difference Method

For example: one-dimensional differential problems, in the scalar unknown function $y(x)$.
Place on the x axis a series of points x_k, and aim at computing the values of the unknown function $y(x)$ only in correspondence to these points. To do this, the derivatives in the differential equation are discretized by replacing them with the incremental ratios, using the values $y(x_k)$ of the unknown function at the points x_k. There are several possible choices to write derivatives in a discrete form, even in the elementary case of 1D 1st-order problems.

$$y'(x_k) \approx \frac{y(x_k + \Delta x) - y(x_k)}{\Delta x}; \Delta x = x_{k+1} - x_k \quad \text{Forward difference}$$

$$y'(x_k) \approx \frac{y(x_k) - y(x_k - \Delta x)}{\Delta x}; \Delta x = x_k - x_{k-1} \quad \text{Backward difference}$$

$$y'(x_k) \approx \frac{y(x_{k+1}) - y(x_{k-1})}{x_{k+1} - x_{k-1}}; \quad \text{Central difference}$$

etc.

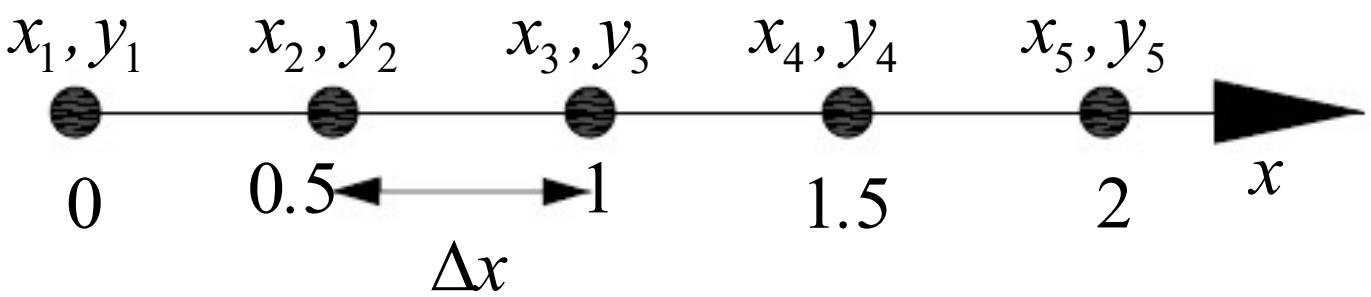
Prescribing now the respect of the discretized differential equation in all the points x_k i.e., in technical terms, "collocating" the equation in the set of all the discrete points x_k, one obtains a system of algebraic equations in the unknowns $y(x_k)$. The system is linear if the starting differential equation was linear; non-linear otherwise.

An elementary example:
a linear first-order equation

$$\begin{cases} y'(x) + y(x) = 0 \\ y(0) = 1 \\ 0 \le x \le 2 \end{cases}$$

$$\text{Analtical solution:} y(x) = e^{-x}$$

Choose to divide the integration interval $0 \le x \le 2$ using a $\Delta x = 0.5$, i.e., using 5 points.



The unknowns are now the values of $y(x)$ in the 5 points adopted to subdivide the integration interval:

$$y_1 = y(x=0)$$
$$y_2 = y(x=0.5)$$
$$y_3 = y(x=1)$$
$$y_4 = y(x=1.5)$$
$$y_5 = y(x=2)$$

Actually, the real unknowns are only 4, since the boundary condition gives us immediately the value $y_1 = y(0) = 1$. Writing now a discretized form of the differential equation in the remaining 4 points one obtains the algebraic system that governs the discretized problem.

We choose to adopt the backward difference

$$y'(x_k) \approx \frac{y(x_k) - y(x_k - \Delta x)}{\Delta x}; \Delta x = x_k - x_{k-1}$$

more convenient here in view of the boundary condition at $x = 0$. The discrete form of the starting equation at each point x_k becomes then:

$$\frac{y(x_k) - y(x_k - \Delta x)}{\Delta x} + y(x_k) = 0$$

which, written at points x_2, x_3, x_4, and x_5, i.e., for $k = 2, 3, 4,$ and 5, gives

$$\frac{y_2 - y_1}{\Delta x} + y_2 = 0; \quad \frac{y_3 - y_2}{\Delta x} + y_3 = 0;$$

$$\frac{y_4 - y_3}{\Delta x} + y_3 = 0; \quad \frac{y_5 - y_4}{\Delta x} + y_4 = 0;$$

which yields the following approximate discrete solution:

$$y_2 = \frac{2}{3}; y_3 = \frac{4}{9}; y_4 = \frac{8}{27}; y_5 = \frac{16}{81};$$

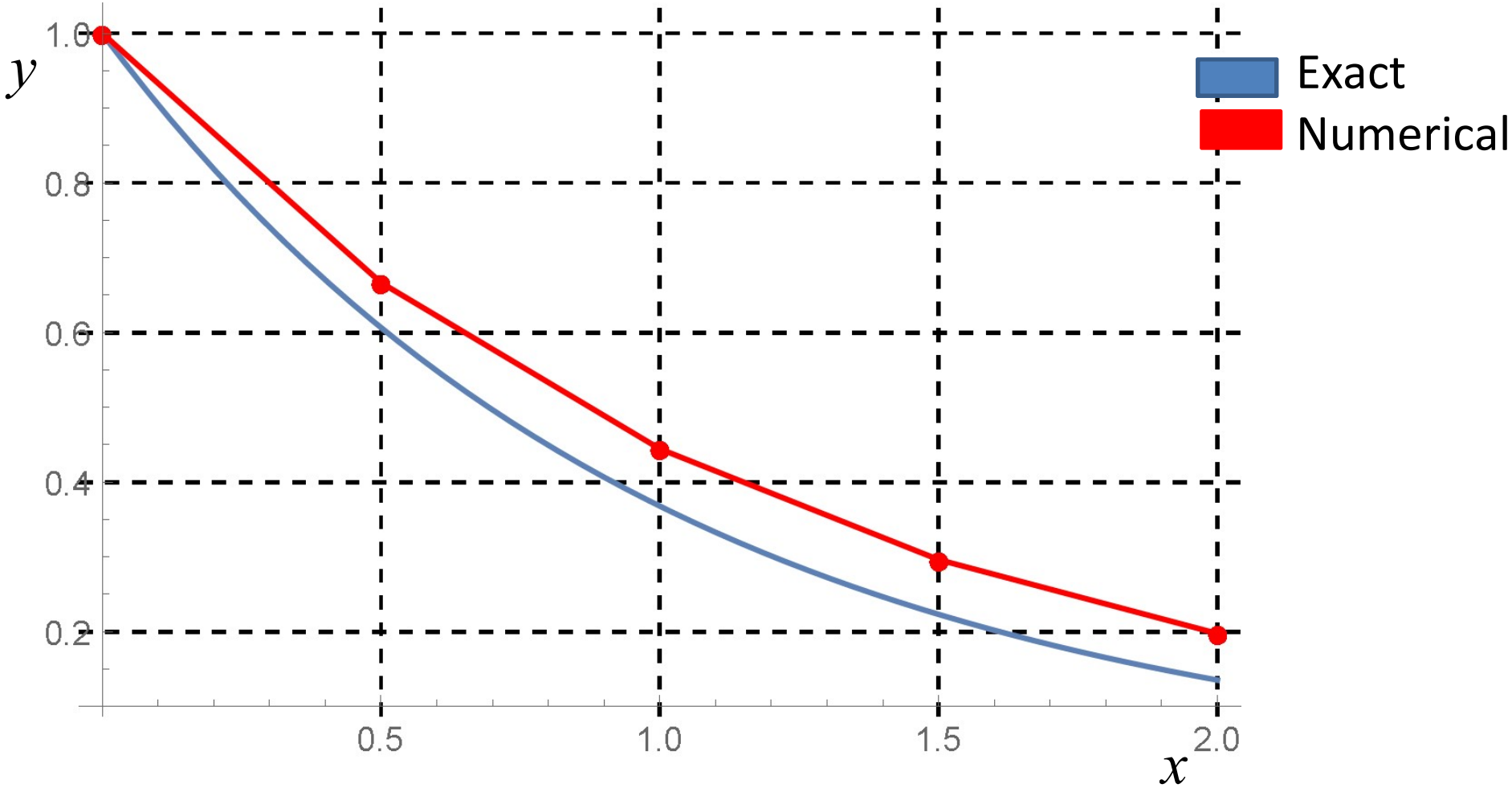
In this extremely simple and special case, the discretized problem is in the form of a so-called difference equation, and admits its own closed-form analytical solution:

$$\frac{y_{n+1} - y_n}{0.5} + y_{n+1} = 0 \quad \text{with } y_1 = 1 \Rightarrow y(x) = \left(\frac{2}{3}\right)^{2x}$$

The exact and approximate results are plotted as follows.

Finte Difference Method

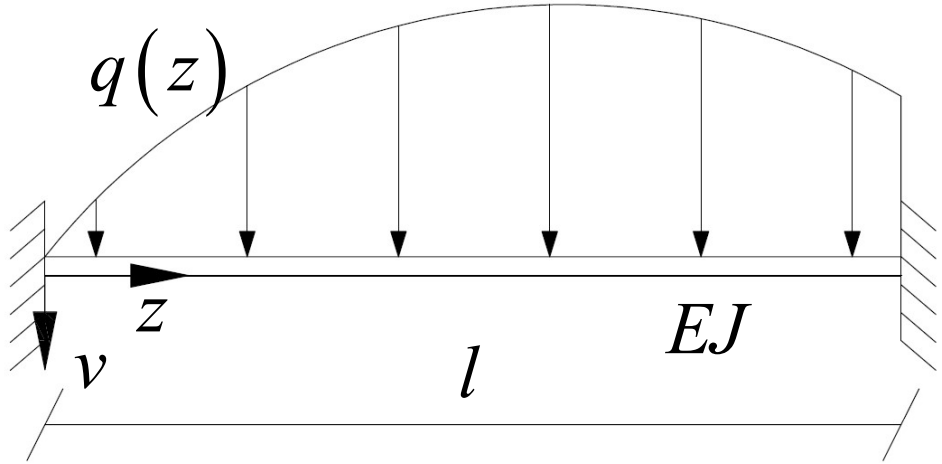
$$\begin{cases} y'(x) + y(x) = 0 \\ y(0) = 1 \end{cases}$$



A more complex example, which shows some difficulties of this method:

$$EJv^{IV}(z) - q(z) = 0$$

$$v(0) = v'(0) = v(l) = v'(l) = 0$$

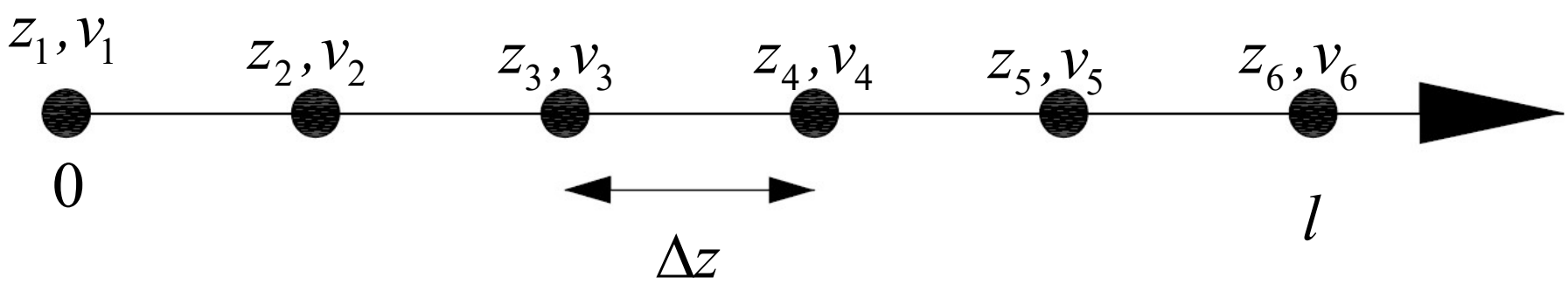


This is the so-called elastica equation (Navier, i.e., equilibrium in terms of displacements only) for a clamped beam subjected to a transversal distributed load q(z). It is a linear ordinary differential equation of the fourth order.

Now we must discretize the space derivatives of *v*(*z*) up to the fourth one. Choose for example the forward difference:

$$v'(z_k) \approx \frac{v(z_k + \Delta z) - v(z_k)}{\Delta z}; \Delta z = z_{k+1} - z_k$$

and choose to subdivide the integration interval $0 \leq z \leq l$ using 6 points:



Owing to the boundary conditions in $z = 0$ and $z = l$, the unknown values of function $v(z)$ are only 4.
In order to discretize the fourth derivative, one starts with the first and second derivatives:

$$v'(z_k) \approx \frac{v(z_k + \Delta z) - v(z_k)}{\Delta z} = \frac{v_{k+1} - v_k}{\Delta z}$$

$$v''(z_k) \approx \frac{v'_{k+1} - v'_k}{\Delta z} = \frac{v_{k+2} - v_{k+1}}{\Delta z^2} - \frac{v_{k+1} - v_k}{\Delta z^2} =$$

$$= \frac{v_{k+2} - 2v_{k+1} + v_k}{\Delta z^2}$$

and proceeding in a similar way for the third and fourth derivatives it is easy to see that

$$v'''(z_k) \approx \frac{v''_{k+1} - v''_k}{\Delta z} \quad \text{contains } v(z_{k+3})$$

$$v^{IV}(z_k) \approx \frac{v'''_{k+1} - v'''_k}{\Delta z} \quad \text{contains } v(z_{k+4})$$

Therefore, for example, the possibly required derivative $v^{IV}(z_5)$ will introduce unknown displacement values up to $v(z_9)$: and obviously, for $z > l$, i.e., beyond the value $v(z_6)$, these are inexistent displacement values.

Nevertheless, some information is given by the boundary conditions on rotations: writing these in discretized form one obtains:

$$v'(z_1) \approx \frac{v_2 - v_1}{\Delta z} = 0 \Rightarrow v_2 = v_1 = 0$$

$$v'(z_6) \approx \frac{v_7 - v_6}{\Delta z} = 0 \Rightarrow \mathrm{v}_7 = v_6 = 0$$

Therefore, the only unknowns to be computed are 3: v_3, v_4, and v_5; and we know the value of the "inexistent" unknown v_7. These considerations, specific of the problem in hand, not general, allow us to compute the 3 unknown displacement values v_3, v_4, and v_5 by "collocating" the discretized differential equation only in the 3 points z_1, z_2, and z_3.
This requires the introduction of the unknown values v_1 up to v_7 only, which are either known a priori (v_1, v_2, v_6, and v_7) or precisely the 3 unknowns to be computed, i.e., v_3, v_4, and v_5.

At this point it should be clear how it is difficult to operate in general terms, i.e., thinking of writing a Finite Difference software which can handle any differential equation with any type of boundary conditions: each problem requires its own special handling, which makes problematic a general treatment.

The Finite Difference Method is nowadays seldom adopted, both because it yields less accurate results, at equal computational cost with other methods, but also because, as seen, it presents difficulties in prescribing in a general way the boundary conditions.It was found by mathematicians that it is possible to obtain more accurate results at the same cost, with a much wider generality of application,applying discretization techniques to the so-called variational formulations, if they exist, or to weak form statements such as the PVW, of the starting differential problem.

As it is seen in the 3rd year class, for quasistatic linear Elasticity one of the most commonly adopted variational formulations (also called minimum principles, or Energy Theorems) is the so-called Theorem of stationarity of the Total Potential Energy (TPE), which corresponds to the Navier equations:

$$TPE\left[u_j\left(x_k\right)\right]=\frac{1}{2}\int_V D_{ijhk}\varepsilon_{ij}\varepsilon_{hk}dV-\int_V b_j u_j dV-\int_{s_f} t_j u_j dS$$

$$\varepsilon_{ij}=\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right)$$

$$u_j\left(x_k\right)=\bar{u}_j\left(x_k\right)\ \text{in}\ S_u$$

$$\boxed{\min_u\left\{TPE\left[u_j\left(x_k\right)\right]\middle|\,compatibility\right\}\Rightarrow u_j^{sol}\left(x_k\right)}$$

To prescribe the stationarity of the Total Potential Energy is fully equivalent to prescribing the equilibrium equations written in terms of displacements only, taking care to automatically satisfy compatibility (Navier).

Recall one of the possible ways of writing the Navier equations for the quasi-static case:

$$\frac{\partial^2 u_i}{\partial x_1^{\,2}} + \frac{\partial^2 u_i}{\partial x_2^{\,2}} + \frac{\partial^2 u_i}{\partial x_3^{\,2}} + \frac{1}{1-2\nu}\frac{\partial^2 u_j}{\partial x_i \partial x_j} + \frac{b_i}{G} = \frac{\rho}{G}\frac{\partial^2 u_i}{\partial t^2} + \frac{\zeta}{G}\frac{\partial u_i}{\partial t} \quad \text{in } V$$

which must be accompanied by the boundary conditions on tractions in S_f , which must also be expressed in terms of the unknown displacement function $u_j(x_k)$.

In quasi-static linear elasticity, the Navier equations are a second-order differential problem in the unknown function $u_j(x_k)$, with boundary conditions.
The associated functional TPE[$u_j(x_k)$] contains at most the first derivatives of function $u_j(x_k)$, and for this reason it is called a class C^0 functional.

As a second example, in a fully analogous way, in the structural theory of beams in generalized variables, for a slender beam under flexure, of length *l* and constant bending stiffness *EI*, for the quasi-static case and in small displacements the Total Potential Energy functional is written as follows:

$$TPE\left[v(z)\right] = \frac{1}{2}\int_0^l EJv''^2(z)dz - \int_0^l q(z)v(z)\,dz$$

(omitting for simplicity any term associated to concentrated loads) in which the unknown function *v*(*z*) (the so-called elastica) must comply a priori with all the geometrical boundary conditions.

Also for beams, of course, the stationarity of the Total Potential Energy functional, with respect to the unknown function *v*(*z*), corresponds to a Navier-type equation with boundary conditions:

$$\min_{v(z)} \left\{ TPE\left[v(z)\right] \middle| compatibility \right\}$$

$$\Downarrow$$

$$\left[E(z)J(z)v''(z)\right]'' - q(z) = 0 + b.c.$$

i.e., an equilibrium equation formulated in terms of displacements only, of the Navier type, in which the elastic stress-strain law is prescribed a priori, and compatibility must be guaranteed a priori : the elastica equation.

By means of standard techniques of Calculus (L. Euler, J. L. Lagrange), for the case of beams it is easy to obtain in closed-form the stationarity equation for this functional expressed in terms of a differential equation plus relevant natural boundary conditions.
It is sufficient to exploit the technique of integration by parts.
Considering a real parameter α, one can write the first variation of functional TPE associated to an arbitrary – but compatible – variation d $v(z)$ of the unknown function:

$$\delta TPE\left[v(z),\delta v(z)\right]=\frac{d}{d\alpha}\left(TPE\left[v(z)+\alpha\delta v(z)\right]\right)\Big|_{\alpha=0}=$$

$$=\int_0^l EJv''(z)\delta v''(z)dz-\int_0^l q(z)\delta v(z)dz$$

Setting equal to zero this expression one already obtains a stationarity condition (which corresponds to the PVW, as we will see later on). But here we wish to proceed further, in order to obtain a differential problem.

In order to obtain both the differential equation and the boundary conditions from the stationarity of the TPE, one must operate on the first integral so as to transform the term $\delta v^2(z)$ in it into a $\delta v(z)$, in such a way that both integrals can be grouped together. This can be done by means of a double integration by parts. Taking *EJ* as constant, one can write:

$$\int_0^l EJv''(z)\delta v''(z)dz = \int_0^l EJv''(z)d\delta v' =$$

$$= \left[EJv''\delta v'\right]_0^l - \int_0^l EJ\delta v'(z)dv'' = \left[EJv''\delta v'\right]_0^l - \int_0^l EJ\delta v'(z)v'''dz =$$

$$= \left[EJv''\delta v'\right]_0^l - \int_0^l EJv'''d\delta v =$$

$$= \left[EJv''\delta v'\right]_0^l - \left[EJv'''\delta v\right]_0^l + \int_0^l EJ\delta v dv''' =$$

$$= \left[EJv''\delta v'\right]_0^l - \left[EJv'''\delta v\right]_0^l + \int_0^l EJv^{IV}\delta v dz$$

thus coming to the following expression:

$$\delta TPE\left[v(z),\delta v(z)\right]=$$

$$=\left[EJv''\delta v'\right]_0^l-\left[EJv'''\delta v\right]_0^l+\int_0^l\left[EJv^{IV}(z)-q(z)\right]\delta v(z)\,dz=0$$

which includes both a definite integral and 4 boundary terms. In the definite integral, the arbitrarity of the variation d *v*(*z*) implies that the integrand function must vanish. This provides us with a differential equation: the elastica equation; the remaining boundary terms, that must vanish independently, prescribe the natural boundary conditions, of the static type only where the geometrical ones do not exist. Where some geometrical boundary conditions exist, they are satisfied a priori, which makes automatically vanish the relevant boundary terms.

The effectiveness of the approach is thus confirmed: the stationarity of the TPE, i.e., the solution of the variational formulation, provides automatically both the corresponding governing differential equation and the necessary associated boundary conditions.

This may be very useful, because there may be problems in which it is “easy” to write a variational formulation but it is not easy to write either the equations or the boundary conditions or both.
In the quasi-static theory of beams, the Navier equation (the elastica equation) is a differential problem of the fourth order in the unknown function $v(z)$, with boundary conditions.
The associated functional TPE[$v(z)$] contains at most the second derivative of the unknown function $v(z)$, and for this reason it is called a class C^1 functional.

For elastic solids or beams, and under quasi-static conditions, the stationarity of the Total Potential Energy coincides with the equation of the Principle of Virtual Work in which compatibility is respected a priori: both are in fact "alternative" methods to prescribe equilibrium.
In fact, for linear elastic continua one can write:

$$TPE\left[u_j\left(x_k\right)\right]=\frac{1}{2}\int_V D_{ijhk}\varepsilon_{ij}\varepsilon_{hk}dV-\int_V b_j u_j dV-\int_{s_f} t_j u_j dS$$

$$\Downarrow$$

$$\delta TPE=\int_V D_{ijhk}\delta\varepsilon_{ij}\varepsilon_{hk}dV-\int_V b_j\delta u_j dV-\int_{s_f} t_j\delta u_j dS=$$

$$=\int_V \sigma_{ij}\delta\varepsilon_{ij}dV-\int_V b_j\delta u_j dV-\int_{s_f} t_j\delta u_j dS=0$$

which, interpreting the (compatible) arbitrary variations as virtual quantities, and assuming zero prescribed displacements (which can be considered if desired), coincides with the Principle of Virtual Work.

Similarly, in the theory of linear elastic beams in generalized variables, one can write:

$$TPE\left[v(z)\right]=\frac{1}{2}\int_0^l EJv''^2(z)dz-\int_0^l q(z)v(z)\,dz$$

$$\Downarrow$$

$$\delta TPE\left[v(z),\delta v(z)\right]=\int_0^l EJv''(z)\delta v''(z)dz-\int_0^l q(z)\delta v(z)\,dz=$$

$$\int_0^l M(z)\delta\chi(z)dz-\int_0^l q(z)\delta v(z)\,dz=0$$

which, interpreting the (compatible) arbitrary variations as virtual quantities, and assuming zero prescribed displacements (which can be considered if desired), coincides with the Principle of Virtual Work.

<u>The Principle of Virtual Work (PVW) can be also seen, independently of its relationship with the stationarity of the TPE, as an independent way to express equilibrium in weak form.</u>

The so-called weak form of any set of differential equations is usually obtained by multiplying the equations by a vector function in order to obtain a scalar quantity, and integrating it over the reference volume.

For the example, of interest here, of a weak form of equilibrium in linear Elastodynamics, this is done starting from the dynamic equilibrium equations:

$$\sigma_{ij,i}\left(x_k,t\right)+b_j\left(x_k,t\right)=\rho\left(x_k\right)\ddot{u}_j\left(x_k,t\right)+\zeta\left(x_k\right)\dot{u}_j\left(x_k,t\right)\ \forall t \text{ in } V;$$

then making them scalar multiplying them by a virtual displacement function u_j^v :

$$\sigma_{ij,i}u_j^v+b_ju_j^V-\rho\ddot{u}_ju_j^V-\zeta\dot{u}_j\mathrm{u}_j^V=0\ \forall t \text{ in } V;$$

and finally integrating the result over the reference volume *V*:

$$\int_V\left(\sigma_{ij,i}u_j^v+b_ju_j^V-\rho\ddot{u}_ju_j^V-\zeta\dot{u}_j\mathrm{u}_j^V\right)dV=0\ \forall t$$

This is an identity if the equilibrium equations are satisfied; if they are not, this already expresses equilibrium in the so-called <u>weak form</u>.

In order to obtain a form corresponding to the PVW some Mathematics is needed, with no further change of the meaning of this equation:

$$\int_V \sigma_{ij,i} u_j^v dV = \int_V \left(\sigma_{ij} u_j^v\right)_{,i} dV - \int_V \sigma_{ij} u_{j,i}^v dV$$

Using the divergence theorem:

$$\int_V \sigma_{ij,i} u_j^v dV = \int_S \sigma_{ij} n_i u_j^v dS - \int_V \sigma_{ij} u_{j,i}^v dV =$$

$$= \int_{S_f} t_j u_j^v dS + \int_{S_u} \sigma_{ij} n_i \overline{u}_j^v dS - \int_V \sigma_{ij} u_{j,i}^v dV$$

The last integral is transformed using the compatibility equations, which are therefore prescribed only at this moment into the weak form statement:

$$\int_V \sigma_{ij} u^v_{j,i} dV = \int_V \sigma_{ij} \frac{1}{2}\left(u^v_{j,i} + u^v_{i,j}\right) dV + \int_V \underbrace{\sigma_{ij} \frac{1}{2}\left(u^v_{j,i} - u^v_{i,j}\right)}_{=0} dV =$$

$$= \int_V \sigma_{ij} \varepsilon^v{}_{ij} dV$$

Inserting these results into the weak form statement, and assuming for simplicity zero displacements on the constrained boundary, one has:

$$\underbrace{\int_V b_j u^v_j dV + \int_{S_f} t_j u^v_j dS}_{W_{ext}} = \underbrace{\int_V \sigma_{ij} \varepsilon^v{}_{ij} dV + \int_V \rho \ddot{u}_j u^v_j dV + \int_V \zeta \dot{u}_j u^v_j dV}_{W_{int}} \quad \forall t$$

which is the customary expression of the PVW holding in the linear Elastodynamics case.

We have thus seen the PVW both as the stationarity condition of the TPE and, directly, as the weak form of equilibrium in which compatibility must be satisfied a priori.

Coming back now to the issue of discretizing the starting continuous problem: both the PVW and the TPE (if this last governs a proper variational formulation) allow one to develop discretization methods which work better, for the same cost, than any other method which discretizes the equations, i.e., the “collocation” methods.

The availability of a functional or of a weak form makes in fact possible the development of space discretization methods often simpler than the Finite Difference Method,
and almost always more effective.

It should be clear, however, that in general it is always possible to discretize a weak form, since any differential equation has one, but that it might be impossible to discretize a TPE, because not all the differential problems admit a variational formulation.

Moreover, it must be observed that the Principle of Virtual Work is much more general than the stationarity of the Total Potential Energy, for two reasons:

1. it holds for any type of constitutive law, whereas the TPE theorem is valid only for elastic materials;

2. the PVW can be written also for the Dynamics case, whereas the TPE can not, since the initial conditions on velocity destroy the mathematical bases for the existence of a variational formulation.

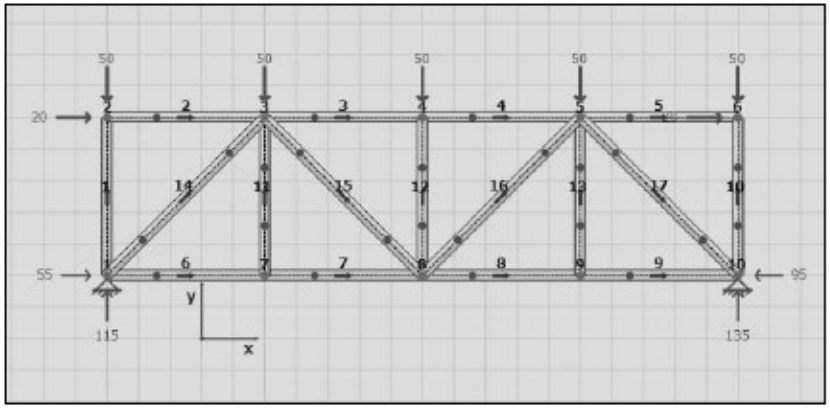
Actually, the PVW is a formidable tool to prescribe equilibrium in a quite general case, if compatibility can be "easily" accounted for.
The use of the PVW becomes essential every time it is not convenient/possible to prescribe equilibrium in a direct form, by means of the differential equations, as in the case when a discretization is introduced.

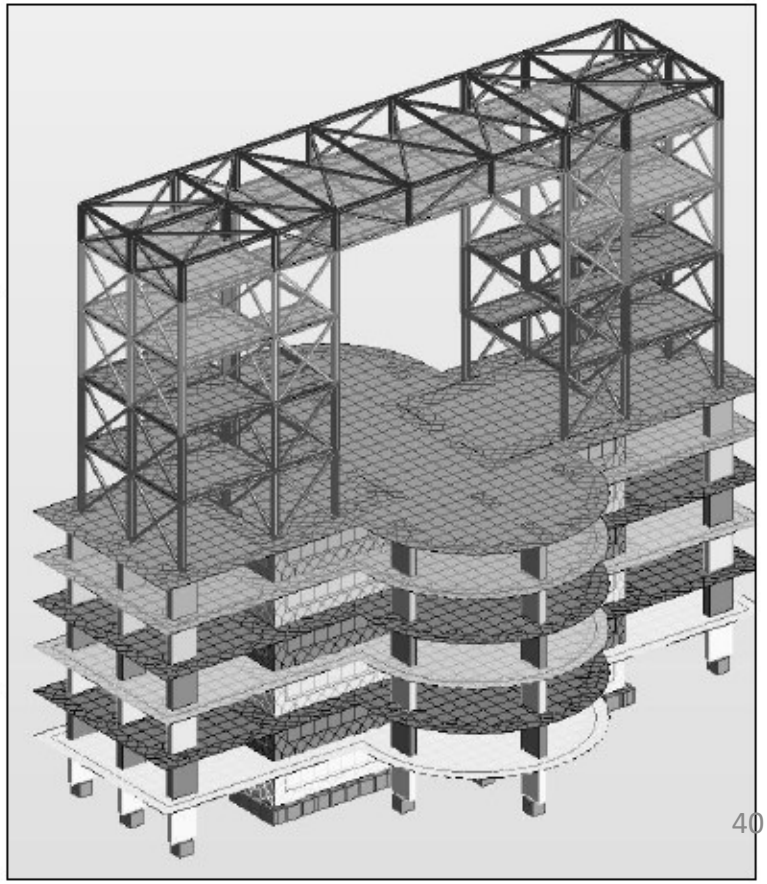
Therefore, since we are interested in discretizing in space the Elastodynamics Problem, we will make reference, as a basis for a discretization technique, mainly to the weak form of the dynamic equilibrium equations, i.e., to the equation of the PVW. Here we will consider, quite briefly, a space discretization methods based on the Principle of Virtual Work: the Finite Element method in Elastodynamics.

The Finite Element Method (FEM)

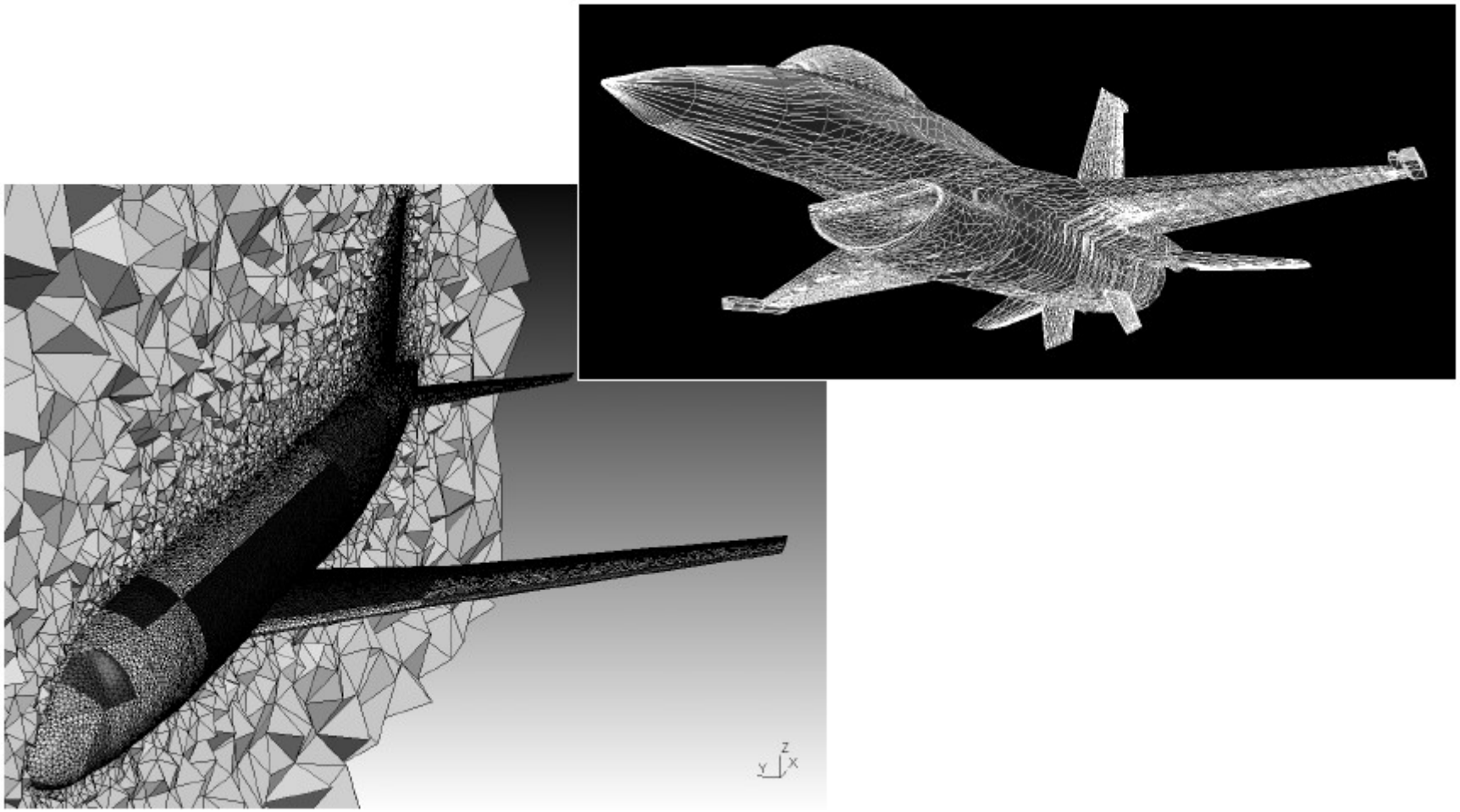
Alexander Hrennikov, 1941; Richard Courant, 1942; Olgierd Zienkiewicz, 1947; John Argyris and Raymond Clough, 1960

Structural elements: Trusses and beams (2D and 3D)

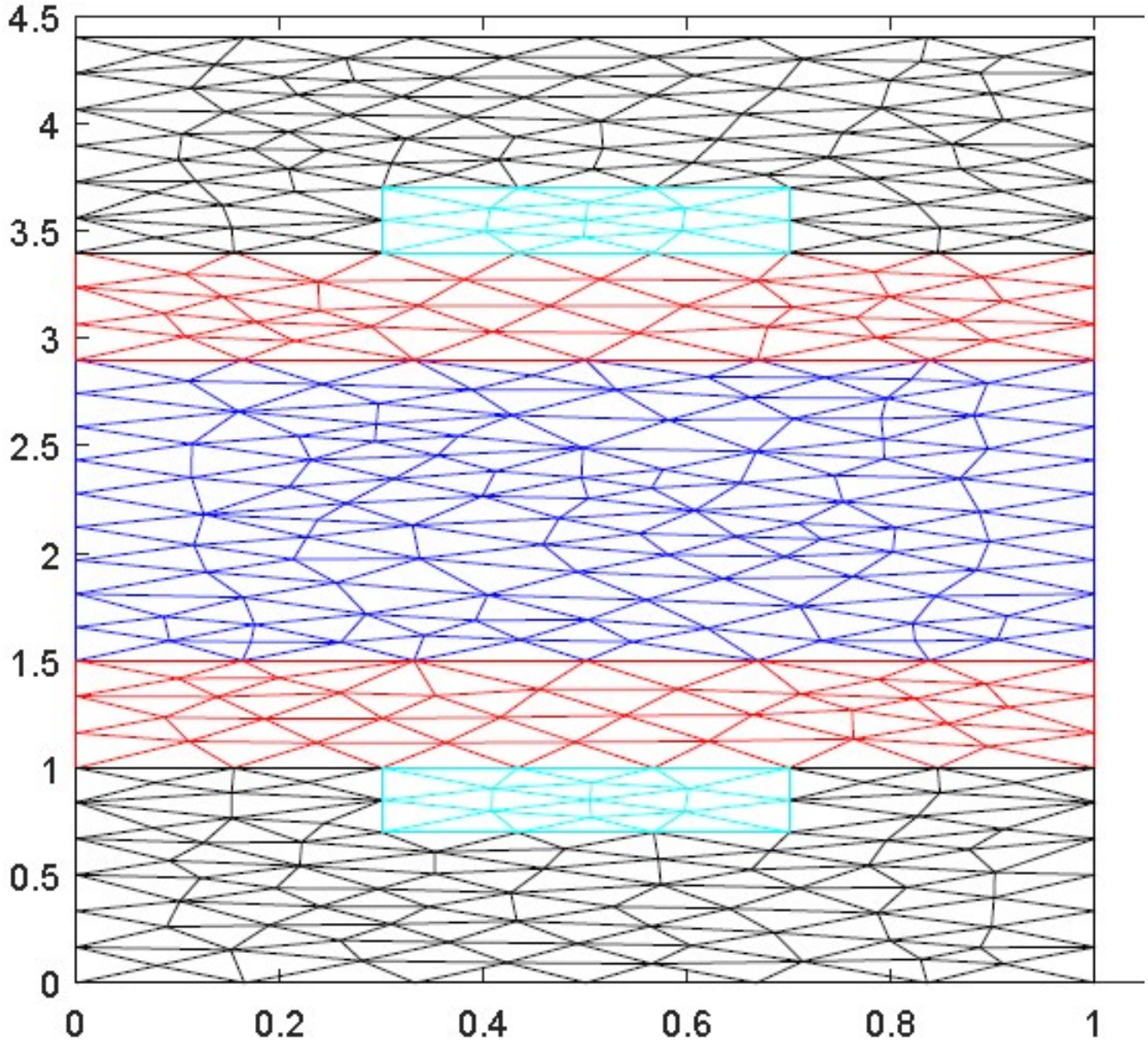




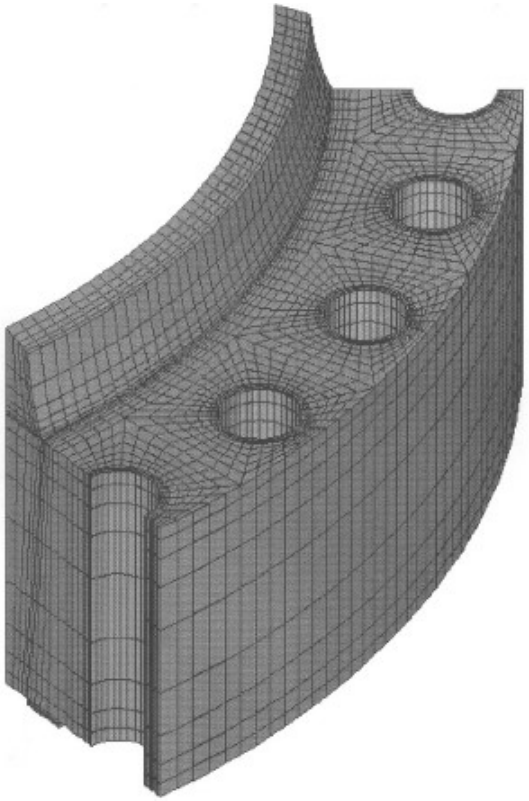
Structural elements: Shells

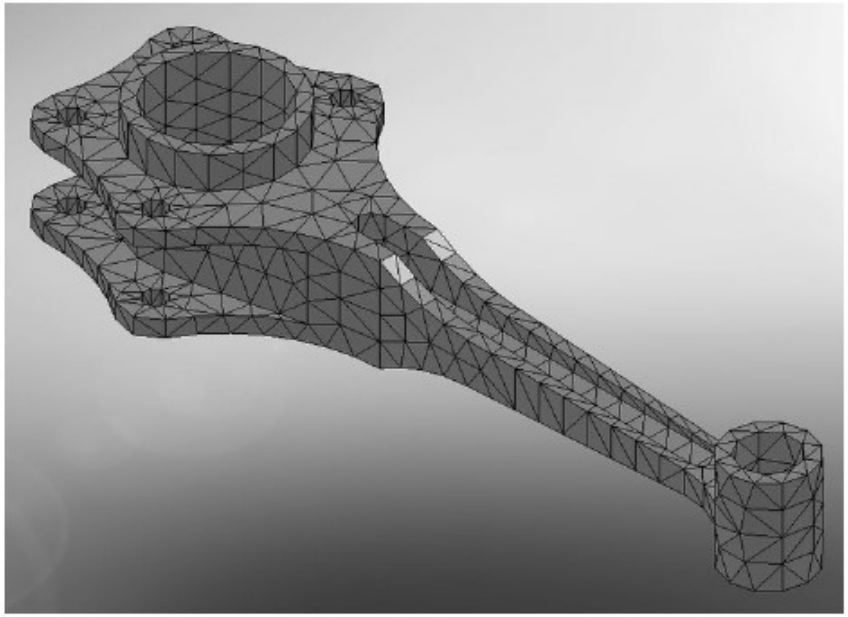


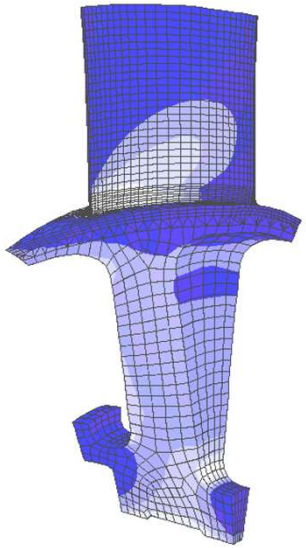
2D continua: plane stress, plane strain, axial symmetry

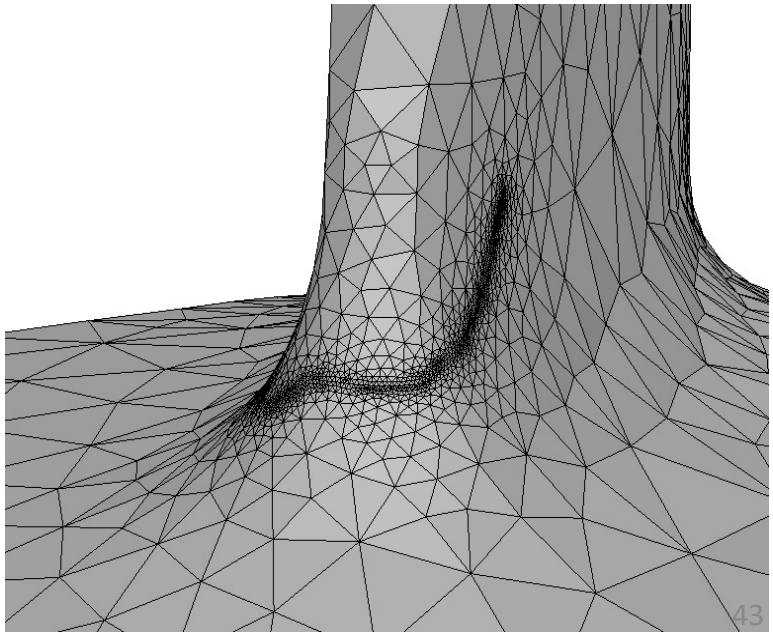


3D continua

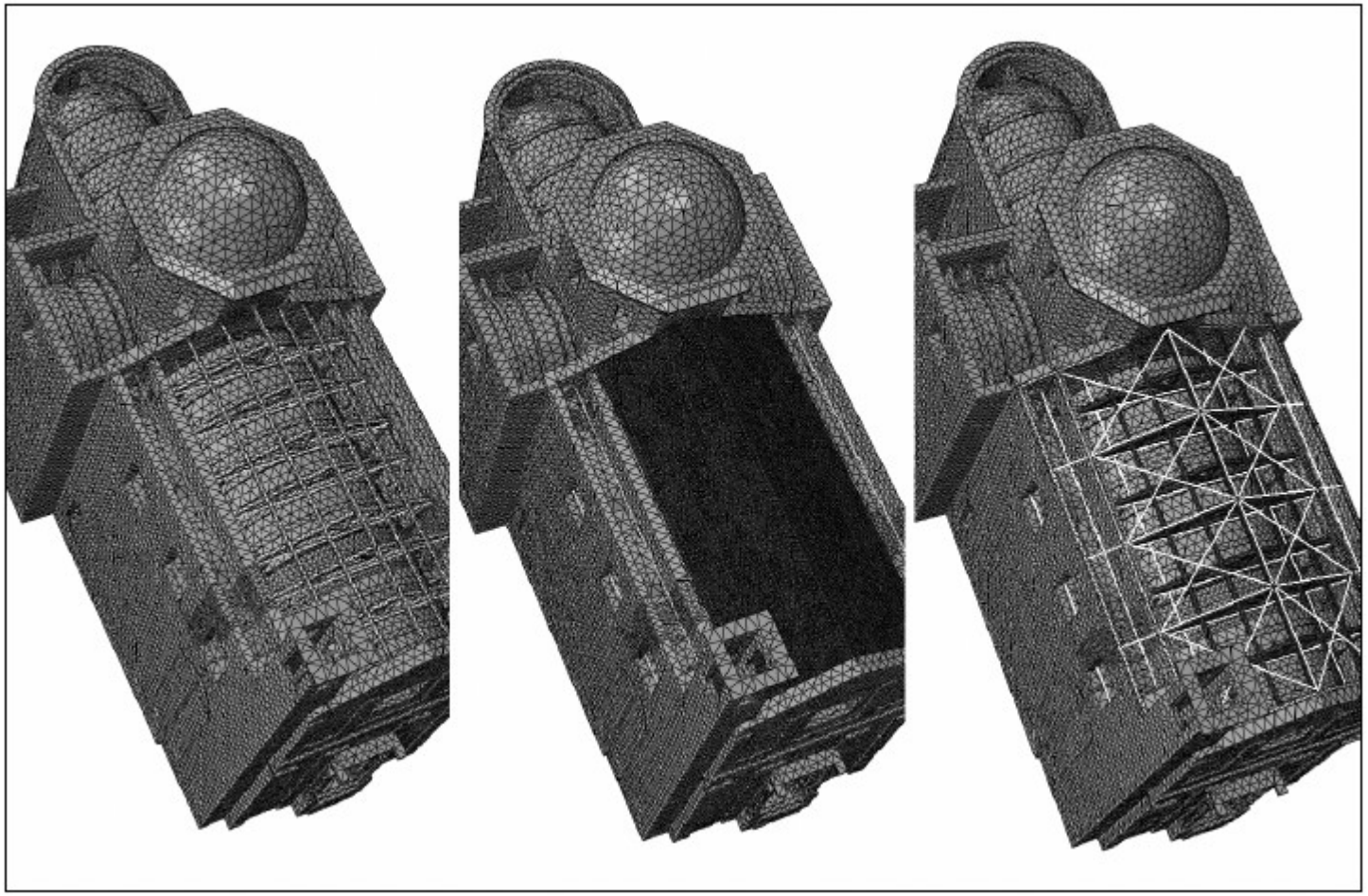








3D continua + structural elements

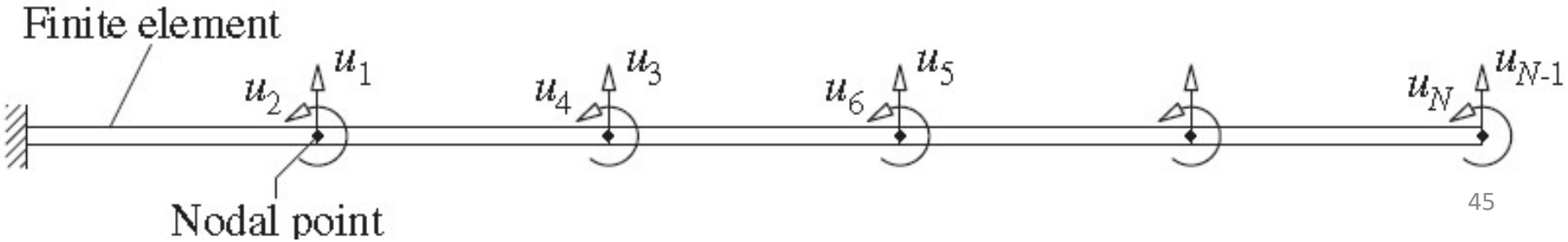


By Professor Francesco Genna

The finite element method is one of the most important developments in applied mechanics.
Although the method is applicable to a wide range of problems, only an introduction is included here with reference to systems that can be idealized as an assemblage of one dimensional structural finite elements.

Finite element approximation

Consider the cantilever beam illustrated in the figure, which is subdivided into a number of segments, called **finite elements**. The set of all the finite elements defines a **mesh**.



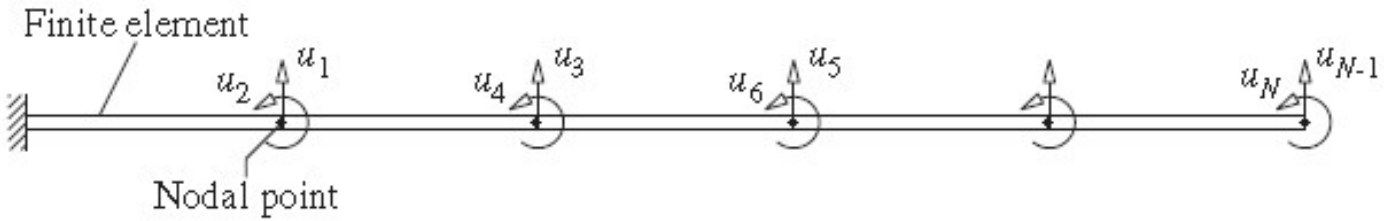
Their size is arbitrary; they may be all of the same size or all different. The elements are interconnected only at **nodes** or **nodal points**. In this simple case the nodal points are the ends of the element, and each node has two DOFs, transverse displacement and rotation. In the FEM the equations of motion are formulated in terms of nodal displacements, which are physically meaningful quantities.

The deflection of the beam is expressed as a linear combination of nodal displacements $u_{i(t)}$ and shape functions $\varphi_{i(x)}$:

$$u(x,t) = \sum_{i=1}^{N_{DOF}} u_i(t)\varphi_i(x)$$

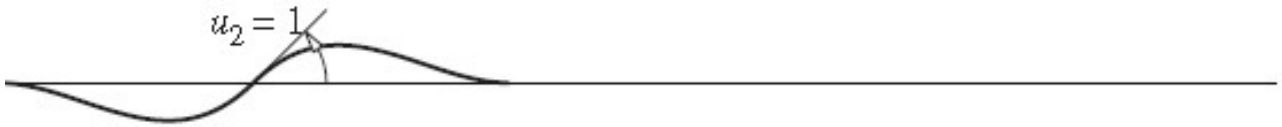
where N_{DOF} is the total number of DOFs in the finite elements mesh

Corresponding to each DOF, a shape function is selected with the following properties: unit value at the DOF and zero value at all the other DOFs; continuous function with continuous first derivative (class C^1 function), linearly independent. Because these functions define the displacements between nodal points, they are also called **interpolation functions**.

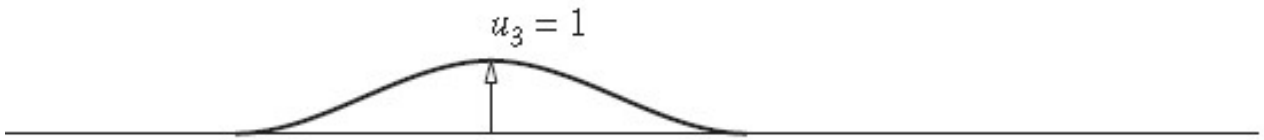


$\varphi_1(x)$

$\varphi_2(x)$



$\varphi_3(x)$



$\varphi_4(x)$

...

The FEM oppers several important advantages. Among them:

1- simple interpolation functions can be chosen for each finite element

2- accuracy of the solution can be improved by increasing the number of finite elements (and, thus, decreasing their size) in the structural idealization (remeshing)

3-the same functions may be chosen for each finite element

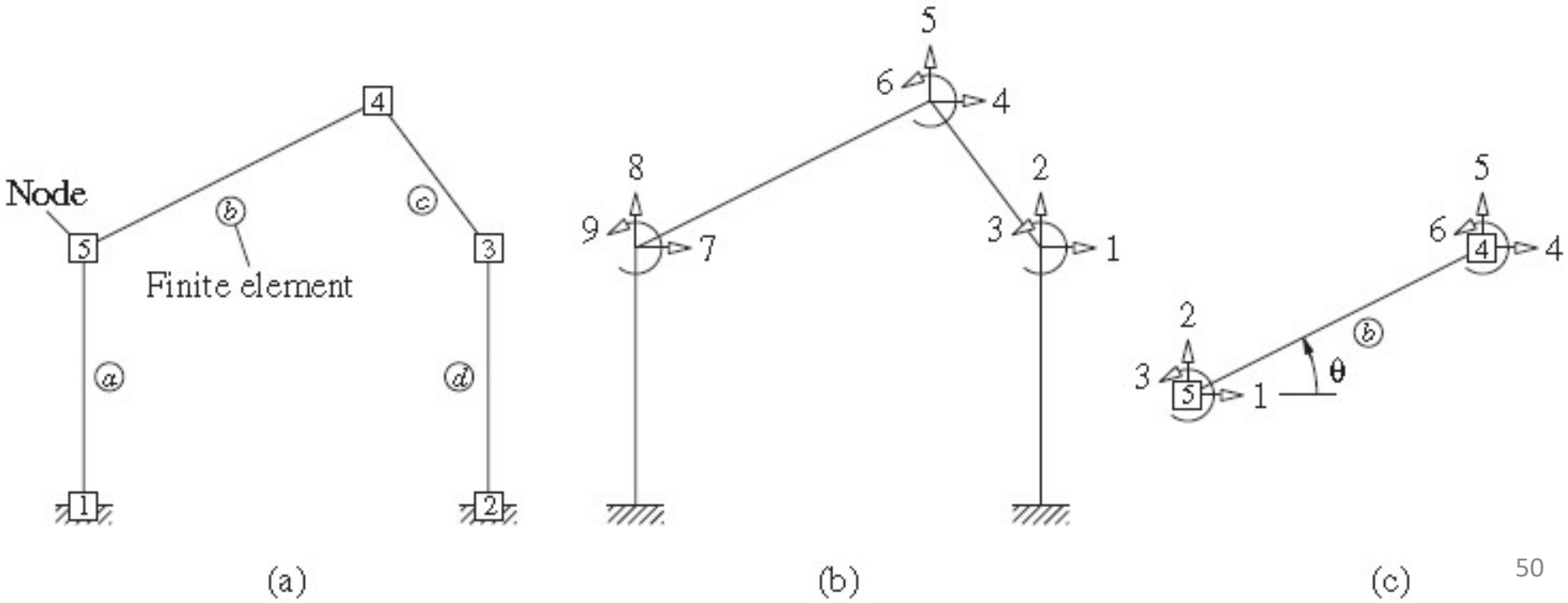
4 structural stiffness and mass matrices are narrowly banded, a property that reduces the computational effort necessary to solve the eqauations of motion

5-nodal unknowns are physically meaningful, since they give the nodal displacements directly

Analysis procedure

The formulation of the equations of motion for a structure by the FEM may be summarized by a sequence of the following steps:

1- idealize the structure as an assemblage of finite elements (mesh) interconnected only at nodes (figure (a)) and define the DOFs at each node (figure (b))



2- for each finite element formthe element stiffness matrix $\mathbf{K}^e$, the element mass matrix $\mathbf{M}^e$, the elemnt damping matrix $\mathbf{C}^e$ and the element applied force vector $\mathbf{F}^e(t)$, with reference to the DOF for the element (figure c)

3-Form the transformation matrix $\mathbf{A}^e$ that relates the displacement vector $\mathbf{u}^e$ and forces $\mathbf{F}^e(t)$ for the element ^{e} to the displacement $\mathbf{u}$ and force $\mathbf{F}(t)$ for the finite element assemblage:

$$\mathbf{u}^e = \mathbf{A}^e\mathbf{u}; \quad \mathbf{F}(t) = \left(\mathbf{A}^e\right)^T \mathbf{F}^e(t); \quad \forall e = 1,\ldots,N_{elements}$$

where $\mathbf{A}^e$ is a boolean matrix consisting of zeros and ones. It simply locates the elements of $\mathbf{K}^e$,$\mathbf{M}^e$,$\mathbf{C}^e$ and $\mathbf{F}^e(t)$ at the proper locations in the global stiffness matrix $\mathbf{K}$, mass matrix $\mathbf{M}$, damping matrix $\mathbf{C}$ and applied force vector $\mathbf{F}(t)$.

4- Assemble the element matrices to the determine the global ones:

$$\mathbf{K} = \mathop{\mathcal{A}}_{e=1}^{N_{elements}} \mathbf{K}^e; \quad \mathbf{M} = \mathop{\mathcal{A}}_{e=1}^{N_{elements}} \mathbf{M}^e;$$

$$\mathbf{C} = \mathop{\mathcal{A}}_{e=1}^{N_{elements}} \mathbf{C}^e; \quad \mathbf{F}(t) = \mathop{\mathcal{A}}_{e=1}^{N_{elements}} \mathbf{F}^e(t);$$

The operator $\mathcal{A}$ denotes the direct assembly procedure for assembling according to matrix $\mathbf{A}^e$,being $N_{elements}$ the total number of elements that are present in the finite element mesh.

5- Formulate the equations of motion for the finite element assemblage:

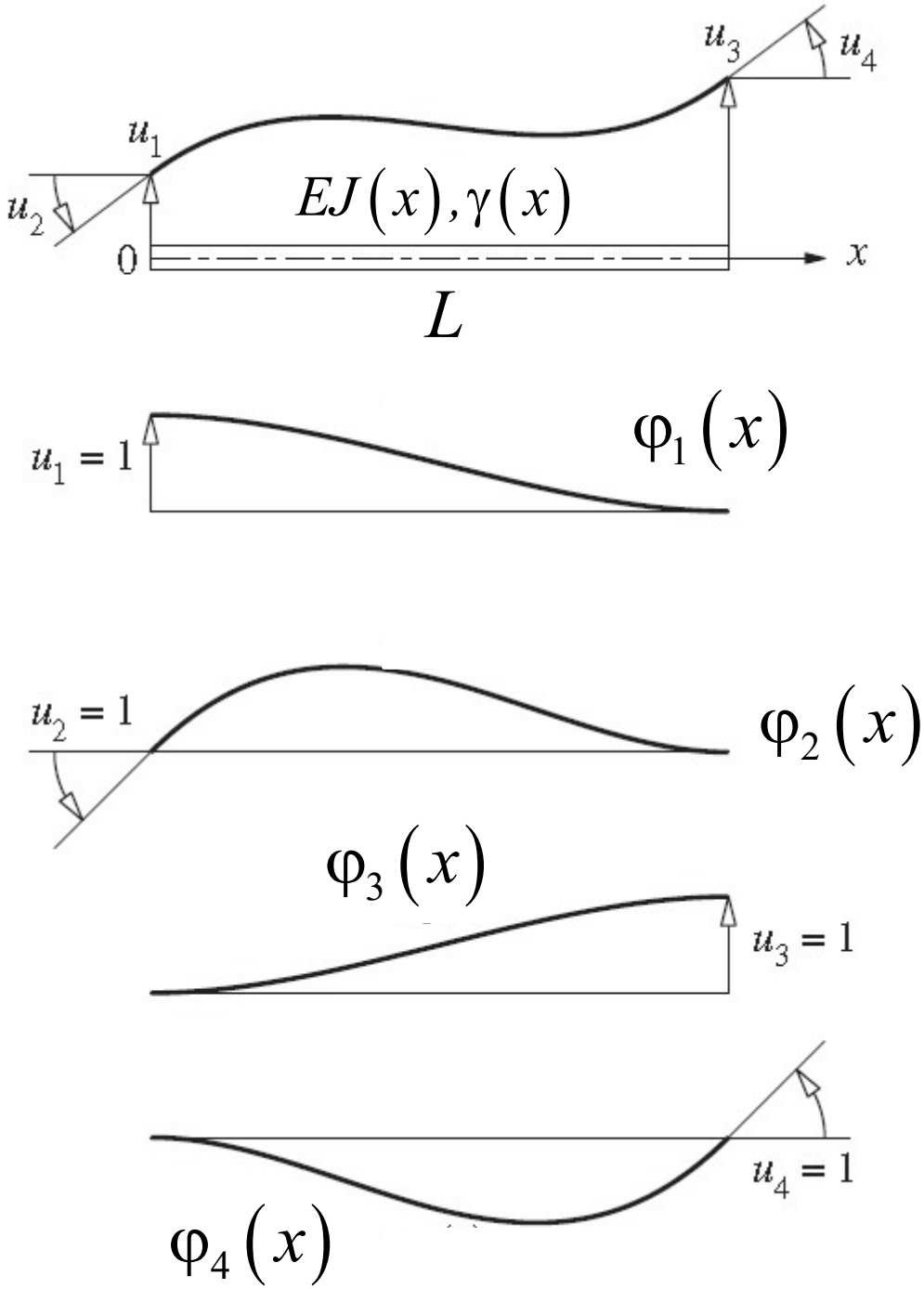
$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{F}(t);$$

The governing equations of motion for a finite element mesh are of the same type as formulated for MDOF systems, but the formulation of element matrices is different.

They can be solved for **u**(t) by the time stepping methods methods that we will see later on.

Element DOFs and shape functions for a 1D beam element

Consider a straight beam elemnt of length L, mass per unit length $\gamma(x)$,and flexural rigidity $EJ(x)$. The two nodes by which the finite element can be assembled into a stucture are located at its ends. If only planar displacements are considered, each node has two DOFs: the transverse displacement and rotation.


u_1
u_2
u_3
u_4
$EJ(x), \gamma(x)$
0
x
L
$u_1 = 1$
$\varphi_1(x)$
$u_2 = 1$
$\varphi_2(x)$
$\varphi_3(x)$
$u_3 = 1$
$u_4 = 1$
$\varphi_4(x)$

The displacement is related to its four DOFs as:

$$u(x,t)=\sum_{i=1}^{4}u_i(t)\varphi_i(x)$$

where the shape function $\varphi_i(x)$ defines the displacement of the element due to unit displacement u_i while constraining other DOFs to zero. Thus $\varphi_i(x)$ satisfies the following boundary conditions:

$$i=1:\quad \varphi_1(0)=1;\varphi'_1(0)=0;\varphi_1(L)=0;\varphi'_1(L)=0;$$

$$i=2:\quad \varphi_2(0)=0;\varphi'_2(0)=1;\varphi_2(L)=0;\varphi'_2(L)=0;$$

$$i=3:\quad \varphi_3(0)=0;\varphi'_3(0)=0;\varphi_3(L)=1;\varphi'_3(L)=0;$$

$$i=4:\quad \varphi_4(0)=0;\varphi'_4(0)=0;\varphi_4(L)=0;\varphi'_4(L)=1;$$

These interpolation functions could be any arbitrary shapes satisfying the boundary conditions. One possibility is the exact deflected shapes of the beam element due to the imposed boundary conditions, but these are difficult to determine if the flexural rigidity varies over the length of the element. However, they can conveniently be obtained for a uniform beam.

Neglecting shear deformations, the equilibrium equation for a uniform beam loaded only at its ends is

$$EJ\frac{d^4u}{dx^4}=0 \quad \text{(elastica equation)}$$

Its general solution is a cubic polynomial:

$$u(x)=a_1+a_2\left(\frac{x}{L}\right)+a_3\left(\frac{x}{L}\right)^2+a_4\left(\frac{x}{L}\right)^3$$

The constants a_i can be determined for each of the four sets of boundary conditions, to obtain

$$\varphi_1(x) = 1 - 3\left(\frac{x}{L}\right)^2 + 2\left(\frac{x}{L}\right)^3;$$

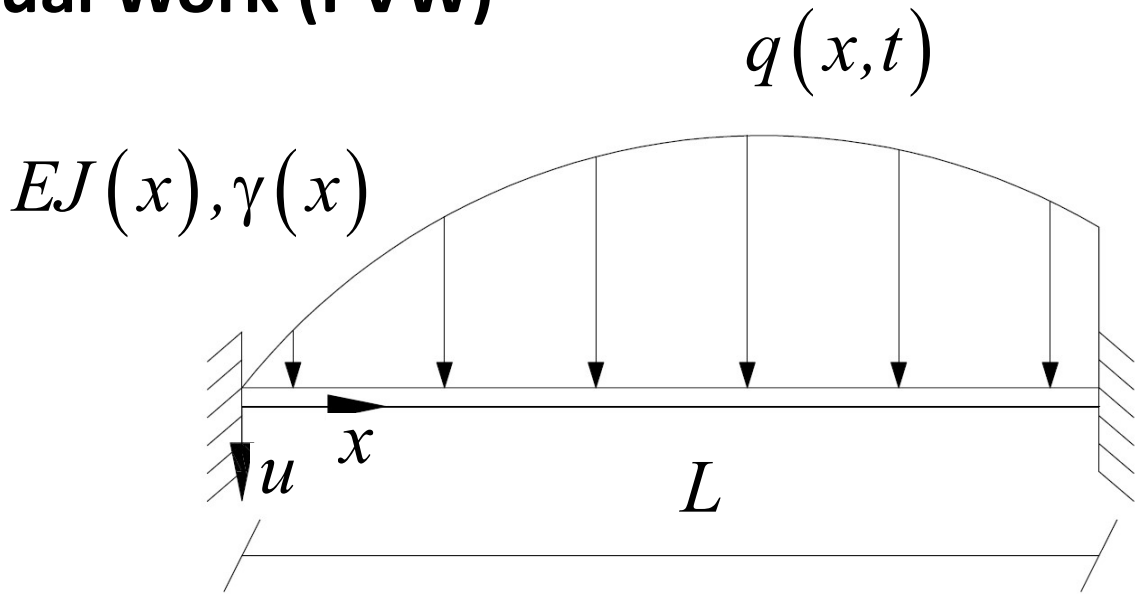
$$\varphi_2(x) = L\left(\frac{x}{L}\right) - 2L\left(\frac{x}{L}\right)^2 + L\left(\frac{x}{L}\right)^3;$$

$$\varphi_3(x) = 3\left(\frac{x}{L}\right)^2 - 2\left(\frac{x}{L}\right)^3;$$

$$\varphi_4(x) = -L\left(\frac{x}{L}\right)^2 + L\left(\frac{x}{L}\right)^3;$$

These interpolation functions can be used to formulate element matrices also for non uniform elements.

Formulation of the element matrices using the Principle of Virtual Work (PVW)



$$W_{ext} = \int_0^L q(x,t)\delta u(x)\,dx$$

$$W_{int} = \int_0^L EJ(x)u''(x,t)\,\delta u''(x)\,dx + \int_0^L \gamma(x)\ddot{u}(x,t)\delta u(x)\,dx +$$

$$+ \underbrace{\int_0^L \zeta\dot{u}(x,t)\delta u(x)\,dx}_{\text{assumed to be vanishing for simplicity}}$$

Inserting finite elemnt approximation both for displacement $u(x,t)$ and for virtual displacement field (which is time-independent), namely

$$u(x,t) = \sum_{i=1}^{N_{DOF}} u_i(t)\varphi_i(x)$$

$$\delta u(x) = \sum_{i=1}^{N_{DOF}} \delta u_i \ \varphi_i(x)$$

And equating the external and the internal works, one obtains

$$\sum_{j=1}^{N_{DOF}} \left\{ \int_0^L EJ(x)\varphi_j''(x)u_j(t)\varphi_i''(x)\,dx\delta u_i + \right.$$

$$\left. + \int_0^L \gamma(x)\varphi_j(x)\ddot{u}_j(t)\varphi_i(x)dx\delta u_i \right\} =$$

$$= \int_0^L q(x,t)\varphi_i(x)\,dx\delta u_i \quad \forall \delta u_i, i = 1,\ldots,N_{DOF}$$

Since the equation has to be valid $\forall \delta u_i$, components of virtual displacement can be simplified, thus obtaining

$$\sum_{j=1}^{N_{DOF}} \left\{ \int_0^L EJ(x) \varphi_j''(x) \varphi_i''(x) dx u_j(t) + \right.$$

$$\left. + \int_0^L \gamma(x) \varphi_j(x) \varphi_i(x) dx \ddot{u}_j(t) \right\} =$$

$$= \int_0^L q(x,t) \varphi_i(x) dx$$

Such an equation can be written in matrix form as

$$\mathbf{K}\mathbf{u}(t) + \mathbf{M}\ddot{\mathbf{u}}(t) = \mathbf{F}(t);$$

where $\mathbf{u}(t)$ is the vector containing nodal unknowns,

K is the **stiffness matrix**, whose components are expressed as

$$\boxed{k_{ij} = \int_0^L EJ(x)\varphi_i''(x)\varphi_j''(x)\,dx}$$

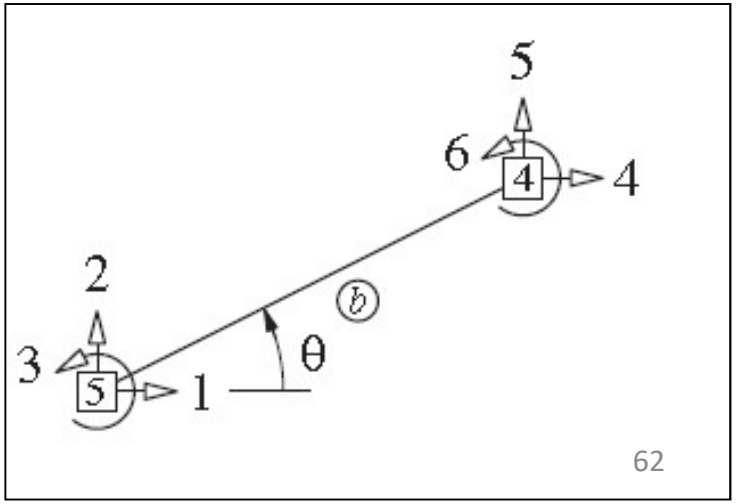
By definition, the stiffness influence coefficient k_{ij} of the beam is the force in DOF i due to unit displacement in the DOF j.

The symmetric form of k_{ij} shows that the stiffness matrix is symmetric.For a uniform finite element with EJ(x)=EJ, the integrals involved in the computation of the stiffness matrix can be evaluated analitically, resulting in the element stiffness matrix

$$\mathbf{K}^e = \frac{EJ}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

These stiffness coefficients are the exact values for a uniform beam, neglecting shear deformations, because the interpolatio functions are the true deflection shapes for this case. For non uniform elements, approximate values for the stiffness coefficients can be determined by numerically evaluating entries k_{ij}. Nevertheless, numerical integration techniques are not tackled in this course.

Before assembling the elemnt stiffness matrices in order to build the global one, note that the 4X4 element stiffness matrix in local element coordinates must be transformed into a 6X6 element matrix in global element coordinates if one has to consider the orientation θ of each finite element .



M is the **mass matrix**, whose components are expressed as

$$\boxed{m_{ij} = \int_0^L \gamma(x)\varphi_i(x)\varphi_j(x)\,dx}$$

The symmetric form of m_{ij} shows that the mass matrix is symmetric.

If we use the same interpolation functions as were used to derve the stiffness matrix, the result obtained is known as the **consistent mass matrix**.

The integrals involved in the computation of entries m_{ij} are evaluated numerically or analitically depending on the function $\gamma(x)$.

For an element with uniform linear density , the integrals can be evaluated analytically to obtain the element consistent mass matrix:

$$\mathbf{M}^e = \frac{\gamma L}{420}\begin{bmatrix} 156 & 22L & 54 & -13L \\ 22L & 4L^2 & 13L & -3L^2 \\ 54 & 13L & 156 & -22L \\ -13L & -3L^2 & -22L & 4L^2 \end{bmatrix}$$

Observe that the consistent mass matrix is not diagonal.
The mass matrix of a finite element can be simplified by assuming that the distributed mass of the element can be lumped as poit masses along the translational DOFs u_1 and u_3 at the ends, with the two masses being determined by static analysis of the beam under its own weight.

For example, if the linear density is γ, a point mass of $\frac{\gamma L}{2}$ will be assigned to each end, leading to

$$\bar{\mathbf{M}}^e = \gamma L \begin{bmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Observe that for a lumped-mass idealization of the finite element, the mass matrix is diagonal.

Diagonal terms associated with rotational DOFs are zero because of the idealization that the mass is lumped in points that have no rotational inertia.

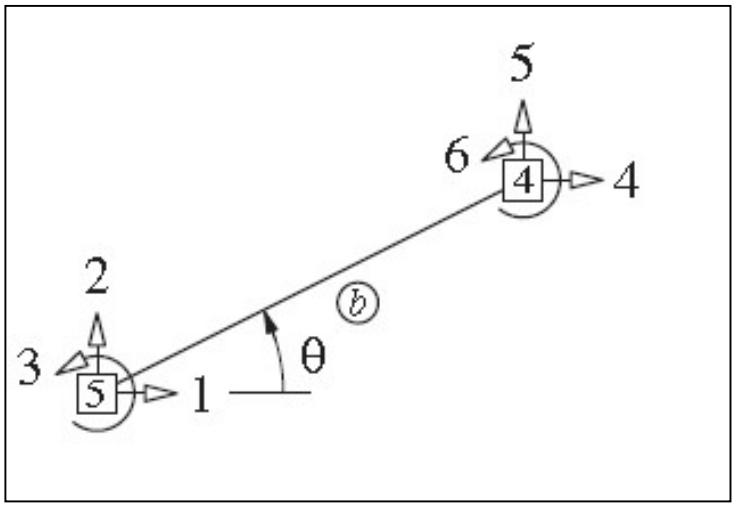
The dynamic analysis of a consistent mass system requires considerably more computational effort than does a lumped mass idealization, for two reasons:

1- the lumped mass matrix is diagonal, whereas the consistent mass matrix has off-diagonal terms

2- the rotational DOFs can be eliminated by static condensation, from the equation of motion for a lumped mass system, whereas all DOFs must be retained in a consistent mass system.

The consistent mass formulation leads to greater accuracy in the results and rapid convergence to the exact results with an increasing number of finite elements, but in practice the improvement is only slight because the inertia forces associated with node rotation are generally not significant in many structural earthquake engineering problems.
The additional computational effort required to achieve a slight increase in accuracy makes the lumped mass idealization a preferable choice.

Also the 4X4 element mass matrix must be transformed into the 6X6 one in global coordinates if one has to consider orientation θ of each finite element in the global coordinate system.



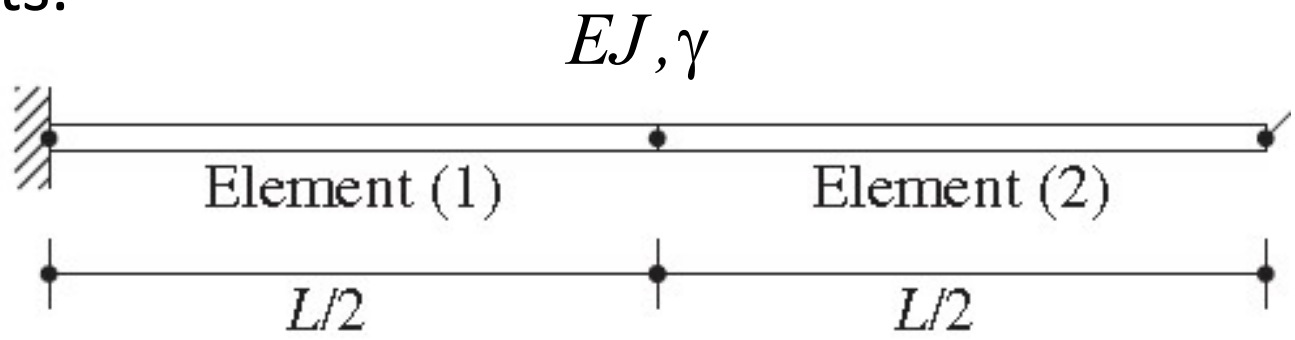
Lastly, **F**(t) is the external force vector, whose components are

$$F_i(t) = \int_0^L q(x,t)\varphi_i(x)dx + \underbrace{\sum_j \bar{F}_j(t)\varphi_i(x_j)}_{\text{this term is present if there are concentrated forces } \bar{F}_j(t) \text{ at locations } x_j}$$

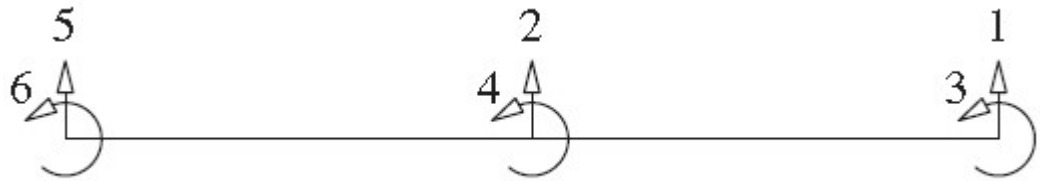
The 4X1 element force vector $\mathbf{F}^e(t)$ in local element coordinates is transformed into the 6X1 in global element coordinates if one considers orientation θ of the element.

Example

Determine the natural frequencies and modes of vibration of a uniform cantilever beam, idealized as an assemblage of two finite elements.



1- The DOFs for the finite element mesh are 6:



DOFs for each finite element, are 4 and are numbered as

2- Form the element stiffness matrices:

Replacing L by L/2 in slide 61, the stiffness matrices for the two finite elements in their local DOFs are given. Since both local element DOFs and global DOFs are defined along the same set of Cartesian coordinates, no transformation of coordinates is required. Therefore:

$$\mathbf{K}^1 = \frac{8EJ}{L^3}\begin{array}{c} \begin{bmatrix} 12 & 3L & -12 & 3L \\ 3L & L^2 & -3L & L^2/2 \\ -12 & -3L & 12 & -3L \\ 3L & L^2/2 & -3L & L^2 \end{bmatrix} \begin{matrix} (5) \\ (6) \\ (2) \\ (4) \end{matrix} \\ \begin{matrix} (5) & (6) & (2) & (4) \end{matrix} \end{array}$$

The number in parentheses alongside rows and columns refer to the global DOFs that correspond to the local DOFs. This information enables local matrices to be assembled into the global ones.

Stiffness matrix for the second element is

$$\mathbf{K}^2 = \frac{8EJ}{L^3}\begin{array}{c} \begin{bmatrix} 12 & 3L & -12 & 3L \\ 3L & L^2 & -3L & L^2/2 \\ -12 & -3L & 12 & -3L \\ 3L & L^2/2 & -3L & L^2 \end{bmatrix} \begin{matrix} (2) \\ (4) \\ (1) \\ (3) \end{matrix} \\ \begin{matrix} (2) & (4) & (1) & (3) \end{matrix} \end{array}$$

3- Assemble the element stiffness matrices:

The stiffness matrix **K** for the finite element mesh is determined by assembling $\mathbf{K}^1$ and $\mathbf{K}^2$ by locating their elements in **K** according to booleanmatrices $\mathbf{A}^1$ and $\mathbf{A}^2$, respectively, where boolean matrices are defined according to

$$\mathbf{u}^1 = \mathbf{A}^1\mathbf{u}; \mathbf{u}^2 = \mathbf{A}^2\mathbf{u}$$

Namely:

$$\mathbf{A}^1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}; \mathbf{A}^2 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix};$$

The global stiffness matrix, thus results

$$\mathbf{K} = \overset{2}{\underset{e=1}{\mathcal{A}}} \mathbf{K}^e = \frac{8EJ}{L^3} \left[\begin{array}{cccc|cc} 12 & -12 & -3L & -3L & 0 & 0 \\ -12 & 24 & 3L & 0 & -12 & -3L \\ -3L & 3L & L^2 & L^2/2 & 0 & 0 \\ -3L & 0 & L^2/2 & 2L^2 & 3L & L^2/2 \\ \hline 0 & -12 & 0 & 3L & 12 & 3L \\ 0 & -3L & 0 & L^2/2 & 3L & L^2 \end{array} \right]$$

Constrained DOFS

Constrained DOFS

4- Form the element mass matrices: replacing L by L/2 in slide 64, gives element mass matrices in their local DOFs. Thus

$$\mathbf{M}^1 = \frac{\gamma L}{840}\begin{array}{c} \begin{bmatrix} 156 & 11L & 54 & -6.5L \\ 11L & L^2 & 6.5L & -0.75L^2 \\ 54 & 6.5L & 156 & -11L \\ -6.5L & -0.75L^2 & -11L & L^2 \end{bmatrix} \begin{matrix} (5) \\ (6) \\ (2) \\ (4) \end{matrix} \\ \begin{matrix} (5) & (6) & (2) & (4) \end{matrix} \end{array}$$

$$\mathbf{M}^2 = \frac{\gamma L}{840}\begin{array}{c} \begin{bmatrix} 156 & 11L & 54 & -6.5L \\ 11L & L^2 & 6.5L & -0.75L^2 \\ 54 & 6.5L & 156 & -11L \\ -6.5L & -0.75L^2 & -11L & L^2 \end{bmatrix} \begin{matrix} (2) \\ (4) \\ (1) \\ (3) \end{matrix} \\ \begin{matrix} (2) & (4) & (1) & (3) \end{matrix} \end{array}$$

5- Assemble the element mass matrices:

In a manner analogous to the stiffness matrix one obtains

$$\mathbf{M} = \mathop{\mathcal{A}}_{e=1}^{2} \mathbf{M}^e =$$

$$= \frac{\gamma L}{840} \left[\begin{array}{cccc|cc} 156 & 54 & -11L & 6.5L & 0 & 0 \\ 54 & 312 & -6.5L & 0 & 54 & 6.5L \\ -11L & -6.5L & L^2 & -0.75L^2 & 0 & 0 \\ 6.5L & 0 & -0.75L^2 & 2L^2 & -6.5L & -0.75L^2 \\ \hline 0 & 54 & 0 & -6.5L & 156 & 11L \\ 0 & 6.5L & 0 & -0.75L^2 & 11L & L^2 \end{array} \right]$$

Constrained DOFS

Constrained DOFS

6- Formulate the equations of motion:

The support conditions must be firstly imposed. For the cantilever beam at hand, $u_5=u_6=0$. Thus, the fifth and the sixth rows and comulmns are deleted from the matrices **K** and **M** to obtain

$$\frac{\gamma L}{840}\begin{bmatrix} 156 & 54 & -11L & 6.5L \\ 54 & 312 & -6.5L & 0 \\ -11L & -6.5L & L^2 & 0.75L^2 \\ 6.5L & 0 & 0.75L^2 & 2L^2 \end{bmatrix}\begin{bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \\ \ddot{u}_3 \\ \ddot{u}_4 \end{bmatrix}+$$

$$+\frac{8EJ}{L^3}\begin{bmatrix} 12 & -12 & -3L & -3L \\ -12 & 24 & 3L & 0 \\ -3L & 3L & L^2 & 0.5L^2 \\ -3L & 0 & 0.5L^2 & 2L^2 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

7- Solve the eigenvalue problem:

The natural frequencies are determined by solving

$$\left[\mathbf{K}-\omega^2\mathbf{M}\right]\mathbf{\Psi}=\mathbf{0}$$

thus obtaining

$$\omega_1=3.51772\sqrt{\frac{EJ}{\gamma L^4}};\omega_2=22.2215\sqrt{\frac{EJ}{\gamma L^4}};$$

$$\omega_3=75.1571\sqrt{\frac{EJ}{\gamma L^4}};\omega_4=218.138\sqrt{\frac{EJ}{\gamma L^4}};$$

Now we want to repeat calculations using the lumped mass approximation. The only change is in formulation of the mass matrix.

Element mass matrices are

$$\mathbf{M}^1 = \frac{\gamma L}{2}\begin{array}{c} \begin{bmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} (5) \\ (6) \\ (2) \\ (4) \end{matrix} \\ \begin{matrix} (5) & (6) & (2) & (4) \end{matrix} \end{array}$$

$$\mathbf{M}^2 = \frac{\gamma L}{2}\begin{array}{c} \begin{bmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} (2) \\ (4) \\ (1) \\ (3) \end{matrix} \\ \begin{matrix} (2) & (4) & (1) & (3) \end{matrix} \end{array}$$

Following the procedure described for the consistent mass matrix, the two element mass matrices are assembled in the global one.

Because the mass associated with the global rotational DOFs u_3 and u_4 is zero, they can be eliminated from the stiffess matrix by static condensation. The resulting equation of motion is

$$\frac{\gamma L}{2}\begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{bmatrix} + \frac{48EJ}{7L^3}\begin{bmatrix} 2 & -5 \\ -5 & 16 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solving the eigenvalue problem one obtains

$$\omega_1 = 3.15623\sqrt{\frac{EJ}{\gamma L^4}}; \omega_2 = 16.2580\sqrt{\frac{EJ}{\gamma L^4}};$$

Natural frequencies of a uniform cantilever beam : Consistent-mass finite element and exact solutions

$$\omega_n = \boxed{\alpha_n}\sqrt{\frac{EJ}{\gamma L^4}}$$

	Number of Finite Elements, N_e					
Mode	1	2	3	4	5	Exact
1	3.53273	3.51772	3.51637	3.51613	3.51606	3.51602
2	34.8069	22.2215	22.1069	22.0602	22.0455	22.0345
3		75.1571	62.4659	62.1749	61.9188	61.6972
4		218.138	140.671	122.657	122.320	120.902
5			264.743	228.137	203.020	199.860
6			527.796	366.390	337.273	298.556
7				580.849	493.264	416.991
8				953.051	715.341	555.165
9					1016.20	713.079
10					1494.88	890.732

Source: R. R. Craig, Jr. and A.J. Kurdila, *Fundamentals of Structural Dynamics*, 2nd ed., Wiley, New York, 2006, Section 14.8.

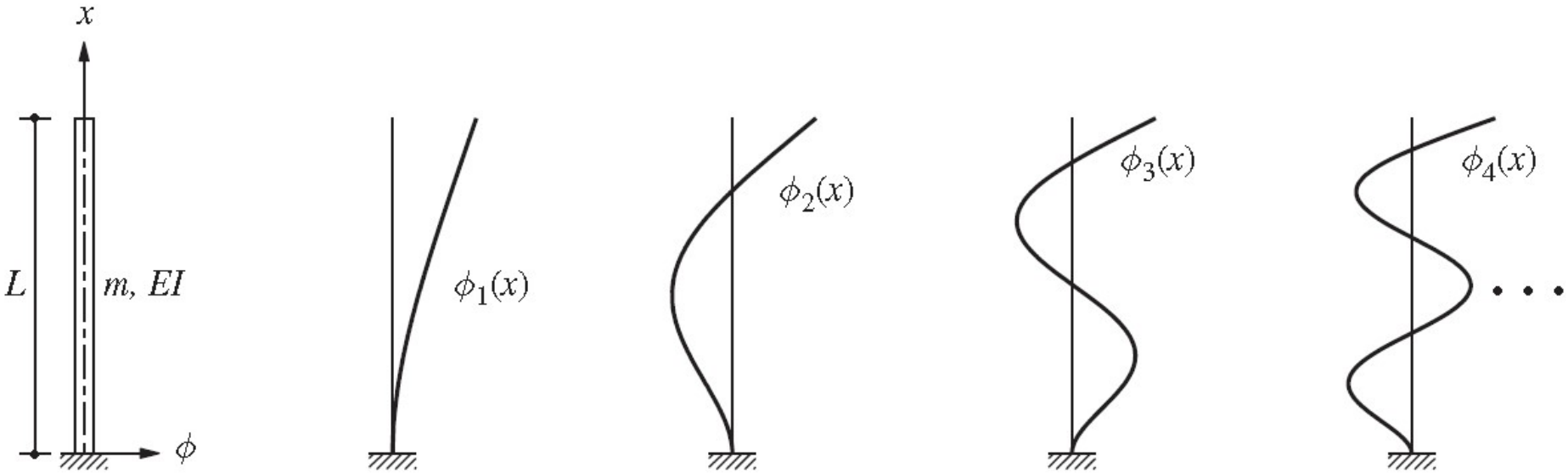
Natural frequencies of a uniform cantilever beam : Lumped-mass finite element and exact solutions

$$\omega_n = \boxed{\alpha_n}\sqrt{\frac{EJ}{\gamma L^4}}$$

	Number of Finite Elements, N_e					
Mode	1	2	3	4	5	Exact
1	2.44949	3.15623	3.34568	3.41804	3.45266	3.51602
2		16.2580	18.8859	20.0904	20.7335	22.0345
3			47.0284	53.2017	55.9529	61.6972
4				92.7302	104.436	120.902
5					153.017	199.860

Source: R. R. Craig, Jr. and A.J. Kurdila, *Fundamentals of Structural Dynamics*, 2nd ed., Wiley, New York, 2006, Section 14.8.

Exact natural vibration modes and frequencies of a uniform cantilever beam



$$\omega_1 = \frac{3.516}{L^2}\sqrt{\frac{EI}{m}} \qquad \omega_2 = \frac{22.03}{L^2}\sqrt{\frac{EI}{m}} \qquad \omega_3 = \frac{61.70}{L^2}\sqrt{\frac{EI}{m}} \qquad \omega_4 = \frac{120.9}{L^2}\sqrt{\frac{EI}{m}}$$

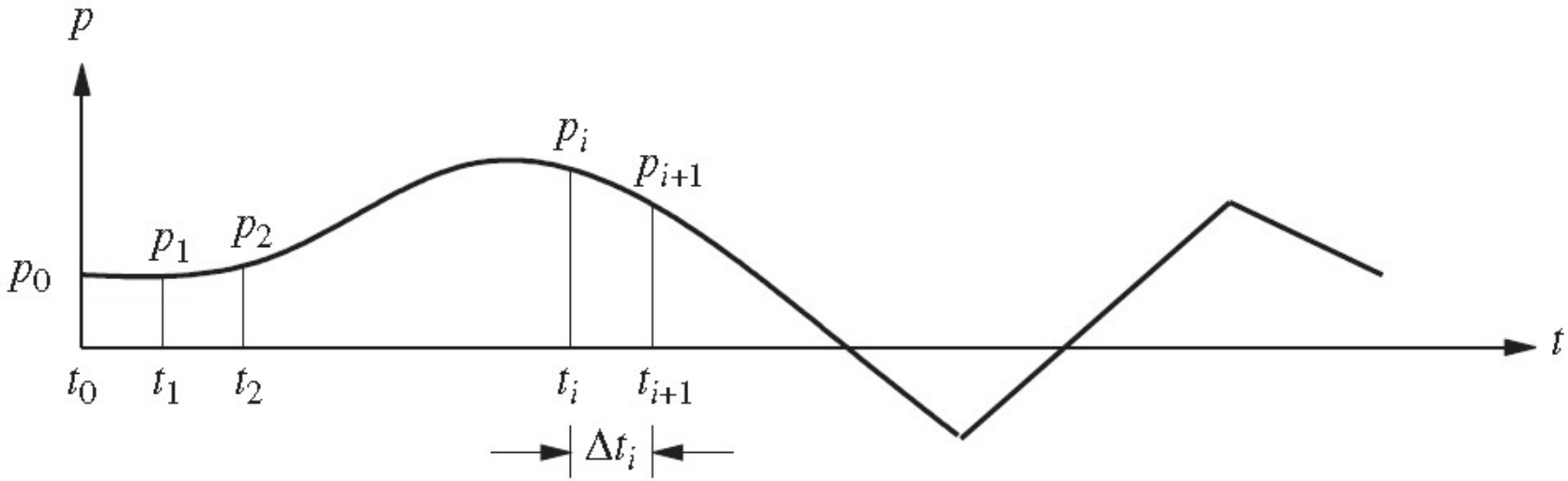
Figure 17.3.2 Chopra book

Time-stepping methods

Analytical solution of the equations of motion for a SDOF system is usually not possible if the excitation varies arbitrarily with time or if the system is non linear. Such problems can be tackled by numerical time-stepping methods for integration of differential equations.

Only a brief presentation of a very few methods that are especially useful in dynamic response analysis of SDOF system is included here in order to provide only the basic concepts underlying these methods and to provide a few computational algorithms.
Furthermore, attention is always restricted to linear systems only.

The external applied force p(t) is given by a set of discrete values $p_i = p(t_i)$, i=0,…,N

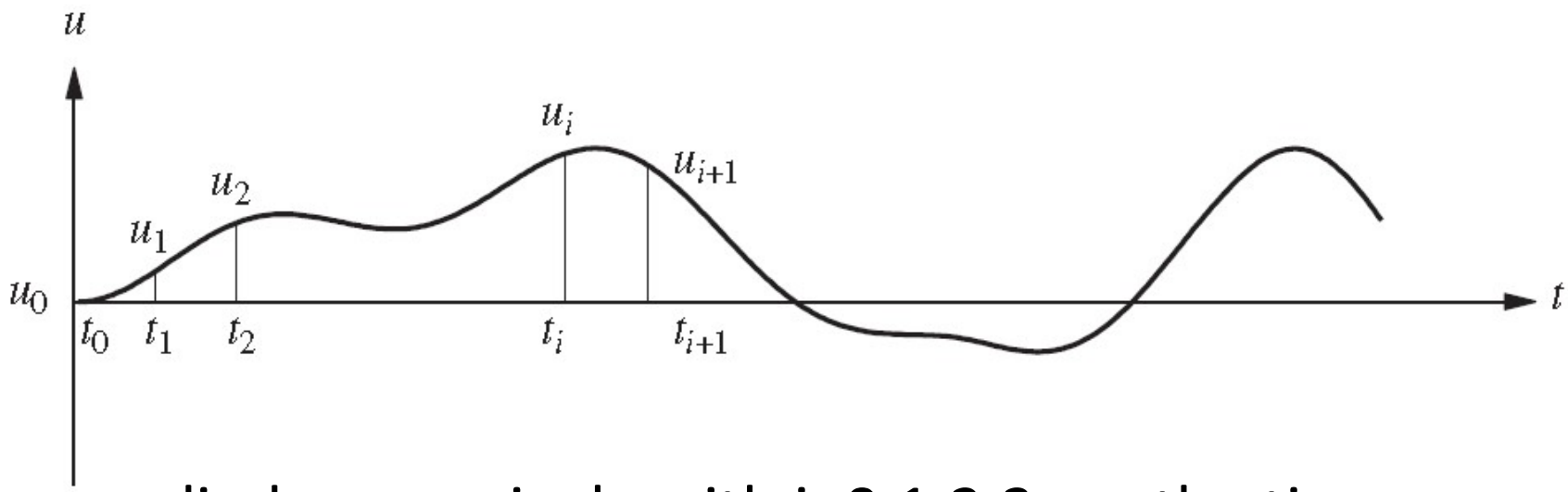


The time interval $\Delta t_i = t_{i+1} - t_i$, is usually taken to be constant, although this is not necessary. The response is determined at the discrete time instants t_i : the displacement u_i, velocity $\dot{u}_i$, and acceleration $\ddot{u}_i$ of the SDOF system. These values, assumed to be known, satisfy at time t_i

$$m\ddot{u}_i + c\dot{u}_i + ku_i = p_i$$

The numerical procedures to be presented will enable us to determine the response quantities $u_{i+1}, \dot{u}_{i+1}, \ddot{u}_{i+1}$ at time t_{i+1} that satisfy

$$m\ddot{u}_{i+1} + c\dot{u}_{i+1} + ku_{i+1} = p_{i+1}$$



When applied successively with i=0,1,2,3,…., the time-stepping procedure gives the desired response at all time instants. The known initial conditions provide the indìformation necessary to start the procedure.

Stepping from time i to i+1 is usually not an exact procedure. Many approximate procedures are possible that are implemented numerically. The three important requirements for a numerical procedure are:

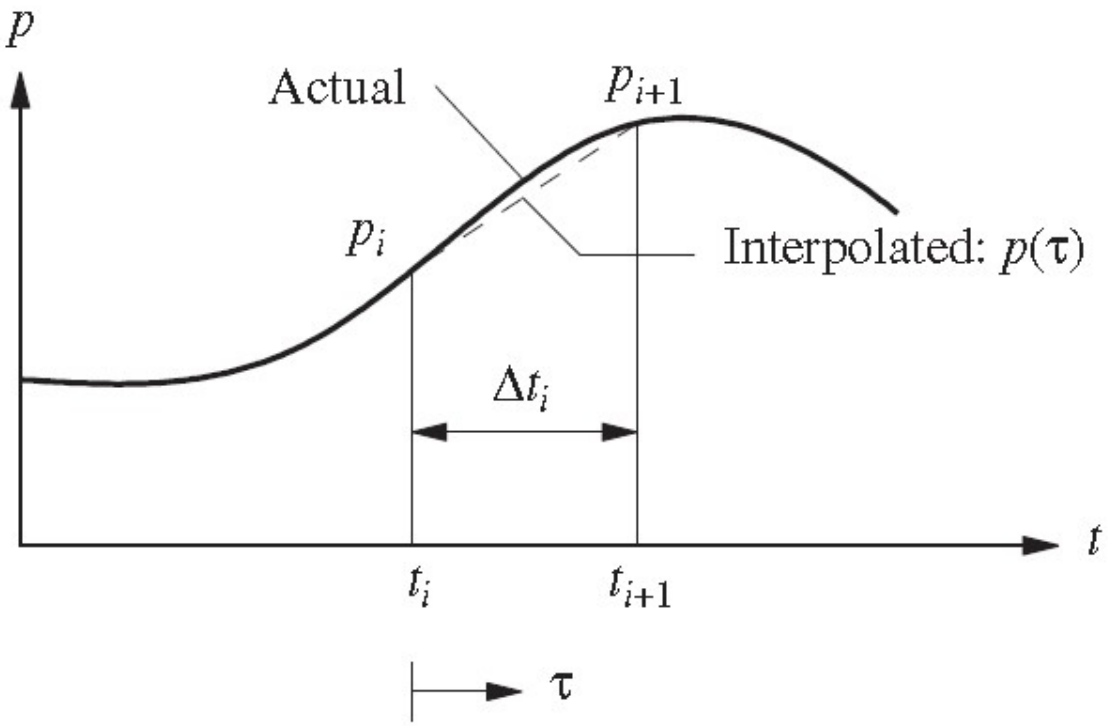
1- **convergence**: as the time step decreases, the numerical solution should approache the exact solution

2- **stability**: the numerical solution should be stable in the presence of numerical round-off errors

3- **accuracy**: the numerical procedure should provide results that are close enough to the exact solution.

1-Methods based on interpolation of excitation

A highly efficient numerical procedure can be developed for linear systems by interpolating the excitation over each time interval and developing the exact solution using the methods studied for SDOF systems. If the time intervals are short, linear interpolation is satisfactory.



Over the time interval $t_i \leq \tau \leq t_{i+1}$, the excitation function is given by

$$p(\tau) = p_i + \frac{\Delta p_i}{\Delta t_i}\tau$$

Where $\Delta p_i = p_{i+1} - p_i$, and $0 \leq \tau \leq \Delta t_i$. For algebraic simplicity, we first consider systems without damping; later the procedure will be extended to include damping.

The equation to be solved is

$$m\ddot{u}(\tau) + ku(\tau) = p_i + \frac{\Delta p_i}{\Delta t_i}\tau$$

The response over the time interval $0 \leq \tau \leq \Delta t_i$ is the sum of three parts:

1- free vibration due to initial displacement u_i and initial velocity $\dot{u}_i$ at $\tau = 0$

2- response to step force p_i with zero initial conditions

3- response to ramp force ${}^{\Delta p_i}/_{\Delta t_i}\,\tau$ with zero initial conditions

Adapting the available solutions for these three cases one has

$$u(\tau) = u_i \cos(\omega\tau) + \frac{\dot{u}_i}{\omega}\sin(\omega\tau) + \frac{p_i}{k}\left(1 - \cos(\omega\tau)\right) +$$

$$+\frac{\Delta p_i}{k}\left(\frac{\tau}{\Delta t_i} - \frac{\sin(\omega\tau)}{\omega\Delta t_i}\right)$$

Differentiating it leads to

$$\frac{\dot{u}(\tau)}{\omega} = -u_i \sin(\omega\tau) + \frac{\dot{u}_i}{\omega}\cos(\omega\tau) + \frac{p_i}{k}\sin(\omega\tau) +$$

$$+\frac{\Delta p_i}{k}\frac{1}{\omega\Delta t_i}\left(1 - \cos(\omega\tau)\right)$$

Evaluating these equations at $\tau = \Delta t_i$ gives the displacement u_{i+1} and velocity $\dot{u}_{i+1}$ at time t_{i+1}.

These equations can be rewritten as recurrence formulas as

$$u_{i+1} = Au_i + B\dot{u}_i + Cp_i + Dp_{i+1}$$
$$\dot{u}_{i+1} = A'u_i + B'\dot{u}_i + C'p_i + D'p_{i+1}$$

Repeating the derivation above for under-critically damped systems (i.e. $\zeta < 1$) shows that recurrence formulas aslo apply to damped systems with the expression for the coefficients A, B, …, D' given in the following table (where ω is denoted as ω_n). They depen on the system parameters ω, k and ζ, and on the time interval $\Delta t \equiv \Delta t_i$.

Since recurrence formulas are derived from exact solution of the equation of motion, the only restriction on the size of the time step Δt is that it permits a close approximation to the excitation function and that it provides response results at closely spaced time intervals so that the response peaks are not missed (**unconditionally stable** method).

Coefficients for recurrence formulas ($\zeta < 1$)

$$A = e^{-\zeta\omega_n \Delta t}\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_D \Delta t + \cos\omega_D \Delta t\right)$$

$$B = e^{-\zeta\omega_n \Delta t}\left(\frac{1}{\omega_D}\sin\omega_D \Delta t\right)$$

$$C = \frac{1}{k}\left\{\frac{2\zeta}{\omega_n \Delta t} + e^{-\zeta\omega_n \Delta t}\left[\left(\frac{1-2\zeta^2}{\omega_D \Delta t} - \frac{\zeta}{\sqrt{1-\zeta^2}}\right)\sin\omega_D \Delta t - \left(1 + \frac{2\zeta}{\omega_n \Delta t}\right)\cos\omega_D \Delta t\right]\right\}$$

$$D = \frac{1}{k}\left[1 - \frac{2\zeta}{\omega_n \Delta t} + e^{-\zeta\omega_n \Delta t}\left(\frac{2\zeta^2-1}{\omega_D \Delta t}\sin\omega_D \Delta t + \frac{2\zeta}{\omega_n \Delta t}\cos\omega_D \Delta t\right)\right]$$

$$A' = -e^{-\zeta\omega_n \Delta t}\left(\frac{\omega_n}{\sqrt{1-\zeta^2}}\sin\omega_D \Delta t\right)$$

$$B' = e^{-\zeta\omega_n \Delta t}\left(\cos\omega_D \Delta t - \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_D \Delta t\right)$$

$$C' = \frac{1}{k}\left\{-\frac{1}{\Delta t} + e^{-\zeta\omega_n \Delta t}\left[\left(\frac{\omega_n}{\sqrt{1-\zeta^2}} + \frac{\zeta}{\Delta t\sqrt{1-\zeta^2}}\right)\sin\omega_D \Delta t + \frac{1}{\Delta t}\cos\omega_D \Delta t\right]\right\}$$

$$D' = \frac{1}{k\,\Delta t}\left[1 - e^{-\zeta\omega_n \Delta t}\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_D \Delta t + \cos\omega_D \Delta t\right)\right]$$

This numerical procedure is especially useful when the excitation is defined at closely spaced time intervals , as for earthquake acceleration, so that the linear interpolation is essentially perfect. If the time step Δt_i is constant, the coefficients A, B, …, D’ need to be computed only once. The exact solution of th equation of motion required in this numerical procedure is feasible only for linear systems. It is conveniently developed for SDOF systems, but would be impractical for MDOF systems unless their response is obtained by the superposition of modal responses.

2-Central difference method

This method is based on a finite difference approximation of the time derivatives of displacement. Taking constant time steps $\Delta t_i = \Delta t$, the central difference expressions for velocity and acceleration at time i are:

$$\dot{u}_i = \frac{u_{i+1} - u_{i-1}}{2\Delta t}; \ddot{u}_i = \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta t^2};$$

Substituting these expressions into the equation of motion at time instant t_i gives

$$m\frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta t^2} + c\frac{u_{i+1} - u_{i-1}}{2\Delta t} + ku_i = p_i$$

where u_i and u_{i-1} are assumed as known from implementation of the procedure at previous time instants.

Transferring these known quantities to the right hand side gives:

$$\left[\frac{m}{\Delta t^2}+\frac{c}{2\Delta t}\right]u_{i+1}=p_i-\left[\frac{m}{\Delta t^2}+\frac{c}{2\Delta t}\right]u_{i-1}-\left[k-\frac{2m}{\Delta t^2}\right]u_i$$

or

$$\hat{k}u_{i+1}=\hat{p}_i,\ \text{with}\ \hat{k}=\left[\frac{m}{\Delta t^2}+\frac{c}{2\Delta t}\right]\ \text{and}$$

$$\hat{p}_i=p_i-\left[\frac{m}{\Delta t^2}+\frac{c}{2\Delta t}\right]u_{i-1}-\left[k-\frac{2m}{\Delta t^2}\right]u_i$$

The unknown u_{i+1} is then given by $u_{i+1}={}^{\hat{p}_i}/_{\hat{k}}$.
The solution u_{i+1} at time i+1 is determined from the equilibrium condition at time i without using the equilibrium condition at time i+1. Such methods are called **explicit methods**.

Observe that known displacements u_i and u_{i-1} are used to compute u_{i+1}. Thus u_0 and u_{-1} are require to determine u_1; the specified initial displecement u_0 is known. To determine u_{-1}, we specialize central difference approximation formulas for i=0 to obtain

$$\dot{u}_0 = \frac{u_1 - u_1}{2\Delta t}; \ddot{u}_0 = \frac{u_1 - 2u_0 + u_{-1}}{\Delta t^2};$$

Solving for u_{-1} from the first equation and substituting in the second gives

$$u_{-1} = u_0 - \Delta t \dot{u}_0 + \frac{\Delta t^2}{2} \ddot{u}_0$$

The initial displacement u_0 and initial velocity $\dot{u}_0$ are given, and the equation of motion at time $t_0 = 0$ provides the initial acceleration:

$$\ddot{u}_0 = \frac{p_0 - c\dot{u}_0 - ku_0}{m}$$

TABLE 5.3.1 CENTRAL DIFFERENCE METHOD†

1.0 *Initial calculations*

1.1 $\ddot{u}_0 = \dfrac{p_0 - c\dot{u}_0 - ku_0}{m}.$

1.2 $u_{-1} = u_0 - \Delta t\, \dot{u}_0 + \dfrac{(\Delta t)^2}{2}\ddot{u}_0.$

1.3 $\hat{k} = \dfrac{m}{(\Delta t)^2} + \dfrac{c}{2\Delta t}.$

1.4 $a = \dfrac{m}{(\Delta t)^2} - \dfrac{c}{2\Delta t}.$

1.5 $b = k - \dfrac{2m}{(\Delta t)^2}.$

2.0 *Calculations for time step i*

2.1 $\hat{p}_i = p_i - au_{i-1} - bu_i.$

2.2 $u_{i+1} = \dfrac{\hat{p}_i}{\hat{k}}.$

2.3 If required: $\dot{u}_i = \dfrac{u_{i+1} - u_{i-1}}{2\Delta t}; \quad \ddot{u}_i = \dfrac{u_{i+1} - 2u_i + u_{i-1}}{(\Delta t)^2}.$

3.0 *Repetition for the next time step*

Replace i by $i + 1$ and repeat steps 2.1, 2.2, and 2.3 for the next time step.

The central difference method will blow up, giving meaningless results, in the presence of numerical round-off if the time step chosen is not short enough. The specific requirement for stability is

$$\frac{\Delta t}{T} < \frac{1}{\pi}$$

where T is the natural time period of the SDOF system at hand. Because of the existence of this limit the method is called **conditionally stable**.

Actually, this limit is never a real constraint for SDOF system because a much smaller time step should be chosen to obtain accurate results. Typically $\Delta t/T < 0.1$, or even shorter, is chosen to define the response accurately.

3- Newmark's method

In 1959, N.M. Newmark developed a family of time-stepping methods based on the following equations:

$$\dot{u}_{i+1} = \dot{u}_i + \left[\left(1-\gamma\right)\Delta t\right]\ddot{u}_i + \gamma\Delta t\ddot{u}_{i+1}$$

$$u_{i+1} = u_i + \Delta t\dot{u}_i + \left[\left(0.5-\beta\right)\Delta t^2\right]\ddot{u}_i + \beta\Delta t^2\ddot{u}_{i+1}$$

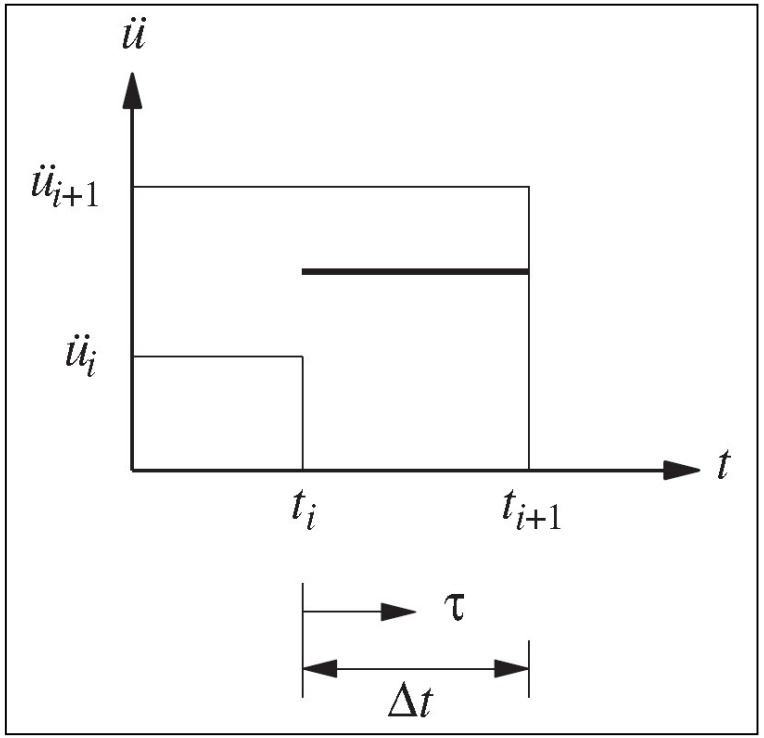
The parameters β and γ define the variation of acceleration over a time step and determine the stability and accuracy characteristics of the method. Typical selection for γ is ½, and $\frac{1}{6} \leq \beta \leq \frac{1}{4}$ is satisfactory from all point of view, including that of accuracy. These two equations, combined with the equilibrium equation at the end of the time step, provide the basis for compunting $u_{i+1}, \dot{u}_{i+1,}\ddot{u}_{i+1}$ at time i+1 from the

known $u_i, \dot{u}_{i,} \ddot{u}_i$ at time i.
Iterations are required to implement these computations because the unknown $\ddot{u}_{i+1}$ appears in the right hand side of equations of the Newmark's family.
For linear systems it is possible to modify Newmark's original formulation, however, to permit solution without iteration.
Before describing this modification, we demonstrate that two special cases of Newmark's method are the well known constant average acceleration and linear acceleration methods.

3.1-Constant average acceleration method

Equation

$$\ddot{u}(\tau) = \frac{1}{2}\left(\ddot{u}_{i+1} + \ddot{u}_i\right)$$



describes the assumption that the variation of acceleration over a time step is constant and equal to the average acceleration. Integration of $\ddot{u}_\tau$ gives

$$\dot{u}(\tau) = \dot{u}_i + \frac{\tau}{2}\left(\ddot{u}_{i+1} + \ddot{u}_i\right)$$

for the variation of the velocity $\dot{u}_\tau$ over the time step. Substituting τ with Δt one has the velocity at time i+1, namely

$$\dot{u}_{i+1} = \dot{u}_i + \frac{\Delta t}{2}\left(\ddot{u}_{i+1} + \ddot{u}_i\right)$$

Integration of $\dot{u}_\tau$ gives

$$u(\tau) = u_i + \dot{u}_i \tau + \frac{\tau^2}{4}\left(\ddot{u}_{i+1} + \ddot{u}_i\right)$$

for the variation $u(\tau)$ of displacement over the time step. Substituting τ with Δt one has the displacement at time i+1, namely

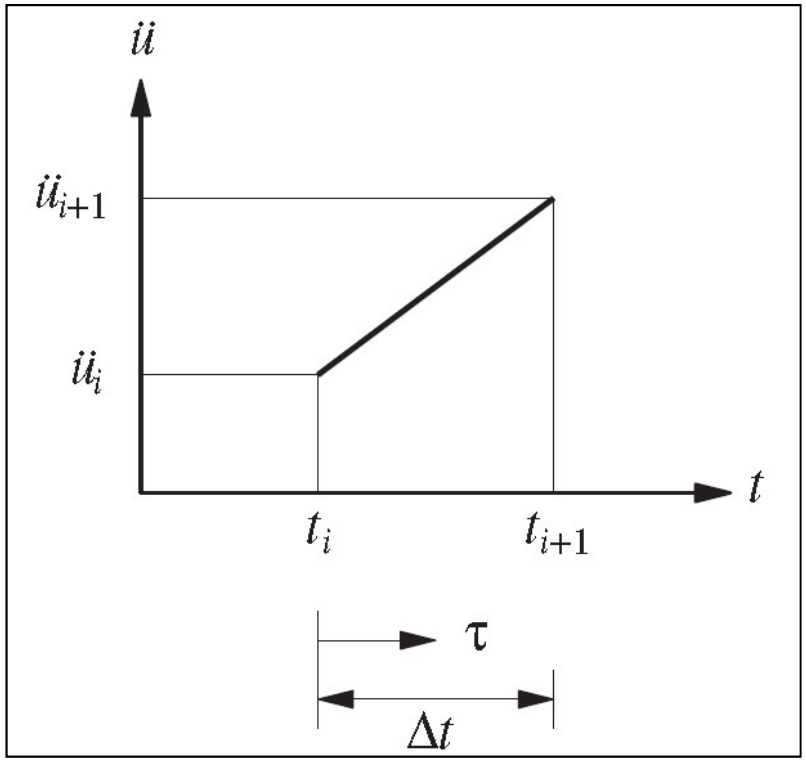
$$u_{i+1} = u_i + \dot{u}_i \Delta t + \frac{\Delta t^2}{4}\left(\ddot{u}_{i+1} + \ddot{u}_i\right)$$

Therefore, Newmark's equation with $\gamma = \frac{1}{2}$ and $\beta = \frac{1}{4}$ are the same as those derived assuming constant average acceleration over the time step.

3.2-Linear acceleration method

Equation

$$\ddot{u}(\tau) = \ddot{u}_i + \frac{\tau}{\Delta t}\left(\ddot{u}_{i+1} - \ddot{u}_i\right)$$



describes the assumption that the variation of acceleration over a time step is linear. Integration of $\ddot{u}_\tau$ gives

$$\dot{u}(\tau) = \dot{u}_i + \ddot{u}_i\tau + \frac{\tau^2}{2\Delta t}\left(\ddot{u}_{i+1} - \ddot{u}_i\right)$$

for the variation of the velocity $\dot{u}_\tau$ over the time step.

Substituting τ with Δt one has the velocity at time i+1, namely

$$\dot{u}_{i+1} = \dot{u}_i + \frac{\Delta t}{2}\left(\ddot{u}_{i+1} + \ddot{u}_i\right)$$

Integration of $\dot{u}_\tau$ gives

$$u(\tau) = u_i + \dot{u}_i \tau + \ddot{u}_i \frac{\tau^2}{2} + \frac{\tau^3}{6\Delta t}\left(\ddot{u}_{i+1} - \ddot{u}_i\right)$$

for the variation $u(\tau)$ of displacement over the time step. Substituting τ with Δt one has the displacement at time i+1, namely

$$u_{i+1} = u_i + \dot{u}_i \Delta t + \Delta t^2 \left(\frac{1}{6}\ddot{u}_{i+1} + \frac{1}{3}\ddot{u}_i\right)$$

Therefore, Newmark's equation with $\gamma = \frac{1}{2}$ and $\beta = \frac{1}{6}$ are the same as those derived assuming linear acceleration over the time step.

For linear systems, it is possible to modify Newmark's original formulation to permit solutions without iteration.
Equation of motion at time i+1 is

$$m\ddot{u}_{i+1} + c\dot{u}_{i+1} + ku_{i+1} = p_{i+1}$$

From the second equation of Newmark's family , $\ddot{u}_{i+1}$ can be expressed in terms of u_{i+1} as

$$\ddot{u}_{i+1} = \frac{1}{\beta\Delta t^2}\left(u_{i+1} - u_i\right) - \frac{1}{\beta\Delta t}\dot{u}_i - \left(\frac{1}{2\beta} - 1\right)\ddot{u}_i$$

Substituting it into the first equation of Newmark's family gives

$$\dot{u}_{i+1} = \frac{\gamma}{\beta\Delta t}\left(u_{i+1} - u_i\right) + \left(1 - \frac{\gamma}{\beta}\right)\dot{u}_i + \Delta t\left(1 - \frac{\gamma}{2\beta}\right)\ddot{u}_i$$

Next, these two last equations are substituted into the governing equation of motion at time i+1, giving

$$\hat{k}u_{i+1} = \hat{p}_{i+1}$$

with

$$\hat{k} = k + \frac{\gamma}{\beta\Delta t}c + \frac{1}{\beta\Delta t^2}m$$

and

$$\hat{p}_{i+1} = p_{i+1} + \left[\frac{1}{\beta\Delta t^2}m + \frac{\gamma}{\beta\Delta t}c\right]u_i + \left[\frac{1}{\beta\Delta t}m + \left(\frac{\gamma}{\beta} - 1\right)c\right]\dot{u}_i +$$

$$+\left[\left(\frac{1}{2\beta} - 1\right)m + \Delta t\left(\frac{\gamma}{2\beta} - 1\right)c\right]\ddot{u}_i$$

With $\hat{k}$ and $\hat{p}_{i+1}$ known from the system properties m, c, and k, algorithm parameters γ and β, and the state of the system at time i defined by $u_i, \dot{u}_{i,} \ddot{u}_i$.

Once $u_{i+1} = \hat{p}_{i+1} / \hat{k}$ is known, the velocity $\dot{u}_{i+1}$ and acceleration $\ddot{u}_{i+1}$ can be computed from

$$\dot{u}_{i+1} = \frac{\gamma}{\beta \Delta t}\left(u_{i+1} - u_i\right) + \left(1 - \frac{\gamma}{\beta}\right)\dot{u}_i + \Delta t\left(1 - \frac{\gamma}{2\beta}\right)\ddot{u}_i$$

and

$$\ddot{u}_{i+1} = \frac{1}{\beta \Delta t^2}\left(u_{i+1} - u_i\right) - \frac{1}{\beta \Delta t}\dot{u}_i - \left(\frac{1}{2\beta} - 1\right)\ddot{u}_i$$

The acceleration can also be obtained from the equation of motion at time i+1 as

$$\ddot{u}_{i+1} = \frac{p_{i+1} - c\dot{u}_{i+1} - ku_{i+1}}{m}$$

Such an equation is needed to obtain $\ddot{u}_0$ to start the time stepping computation. In Newmark's method, the solution at time i+1 is determined from equilibrium condition at time i+1 and, for this reason, it belongs to the class of **implicit methods**.Newmark's method is a stable if

$$\frac{\Delta t}{T} \leq \frac{1}{\pi\sqrt{2}} \frac{1}{\sqrt{\gamma - 2\beta}}$$

with T the natural period of the SDOF system.

For the constant average acceleration method ($\gamma = \frac{1}{2}, \beta = \frac{1}{4}$) this condition becomes

$$\frac{\Delta t}{T} \leq \infty$$

This implies that the constant average acceleration method is **unconditionally stable.**

However, it is accurate only if Δt is small enough to guarantee accuracy.

For the linear acceleration method ($\gamma = \frac{1}{2}, \beta = \frac{1}{6}$), stability limit becomes

$$\frac{\Delta t}{T} \leq 0.551$$

And, for this reason, the method is called **conditionally stable.**

However, as in the case of central difference method, this condition has little significance in the analysis of SDOD systems because a much shorter time step than 0.551 T must be used to obtain an accurate representation of the excitation and response. The following table summarizes the time stepping solution using Newmark's method as it might be implemented in a computer.

TABLE 5.4.2 NEWMARK'S METHOD: LINEAR SYSTEMS†

Special cases

(1) Constant average acceleration method ($\gamma = \frac{1}{2}, \beta = \frac{1}{4}$)

(2) Linear acceleration method ($\gamma = \frac{1}{2}, \beta = \frac{1}{6}$)

1.0 *Initial calculations*

1.1 $\ddot{u}_0 = \dfrac{p_0 - c\dot{u}_0 - ku_0}{m}$.

1.2 Select Δt.

1.3 $a_1 = \dfrac{1}{\beta(\Delta t)^2}m + \dfrac{\gamma}{\beta \Delta t}c; \quad a_2 = \dfrac{1}{\beta \Delta t}m + \left(\dfrac{\gamma}{\beta} - 1\right)c;$ and

$$a_3 = \left(\frac{1}{2\beta} - 1\right)m + \Delta t\left(\frac{\gamma}{2\beta} - 1\right)c.$$

1.4 $\hat{k} = k + a_1$.

2.0 *Calculations for each time step*, $i = 0, 1, 2, \ldots$

2.1 $\hat{p}_{i+1} = p_{i+1} + a_1 u_i + a_2 \dot{u}_i + a_3 \ddot{u}_i$.

2.2 $u_{i+1} = \dfrac{\hat{p}_{i+1}}{\hat{k}}$.

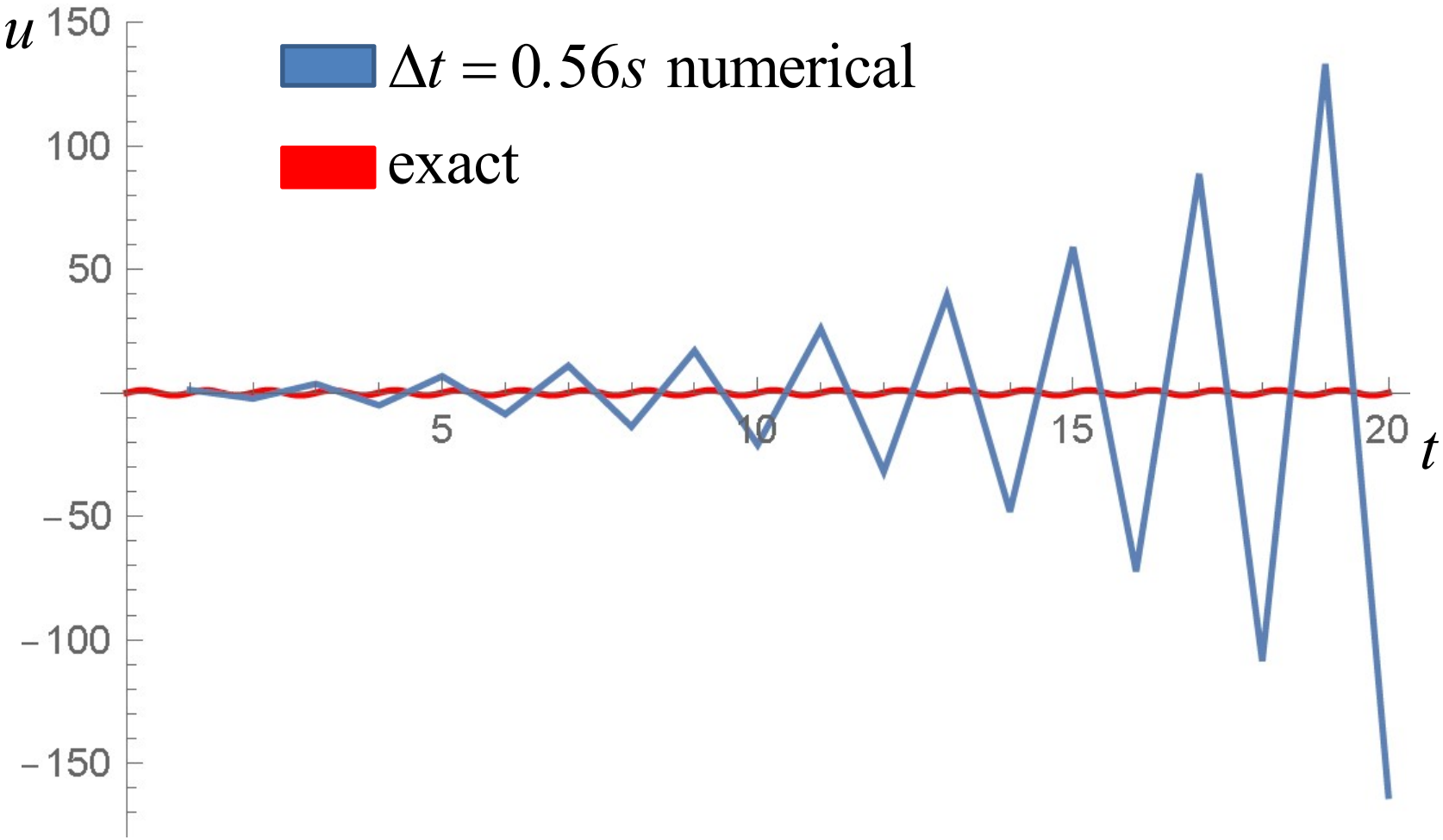
2.3 $\dot{u}_{i+1} = \dfrac{\gamma}{\beta \Delta t}(u_{i+1} - u_i) + \left(1 - \dfrac{\gamma}{\beta}\right)\dot{u}_i + \Delta t\left(1 - \dfrac{\gamma}{2\beta}\right)\ddot{u}_i$.

2.4 $\ddot{u}_{i+1} = \dfrac{1}{\beta(\Delta t)^2}(u_{i+1} - u_i) - \dfrac{1}{\beta \Delta t}\dot{u}_i - \left(\dfrac{1}{2\beta} - 1\right)\ddot{u}_i$.

3.0 *Repetition for the next time step*. Replace i by $i + 1$ and implement steps 2.1 to 2.4 for the next time step.

Free vibration for an undamped SDOF system: Linear acceleration method-example of numerical instability

$$u_0 = 0; \dot{u}_0 = 2\pi; \omega = 2\pi; k = 10; m = k / \left(4\pi^2\right); \zeta = 0;$$



Computational error

Error is inherent an any numerical solution. We do not discuss error analysis from a mathematical point of view. Rather, we examine two important characteristics of numerical solutions to develop a feel for the nature of the errors, and then mention a simple, useful way of managing error.

Consider the free vibration problem

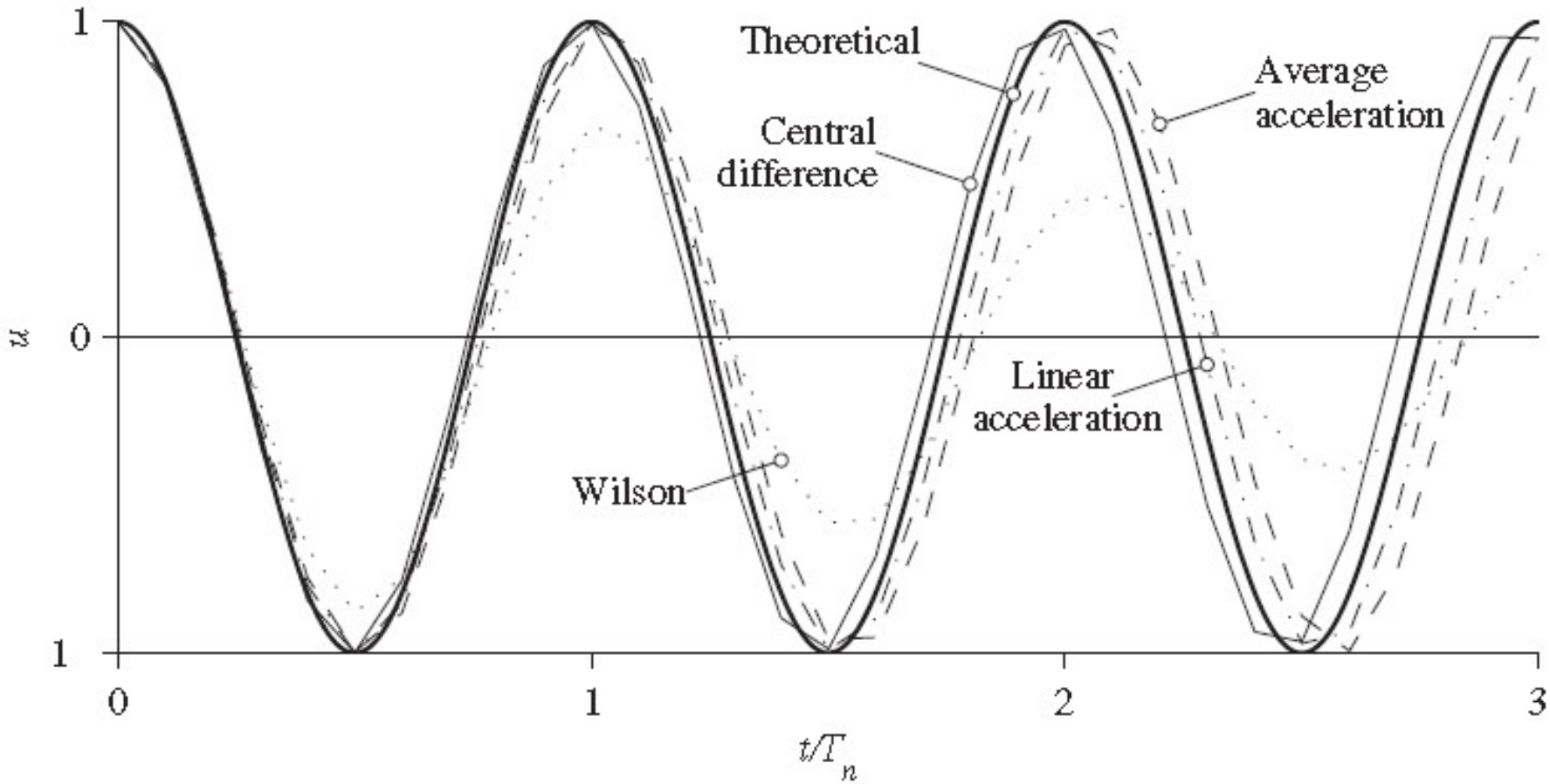
$$m\ddot{u} + ku = 0, \quad u(0) = 1 \text{ and } \dot{u}(0) = 0$$

for which the theoretical solution is

$$u(t) = cos(\omega t)$$

This problem is solved by four numerical methods. The numerical results obtained using $\Delta t = 0.1T$ are compared with the theoretical solution in the following figure.

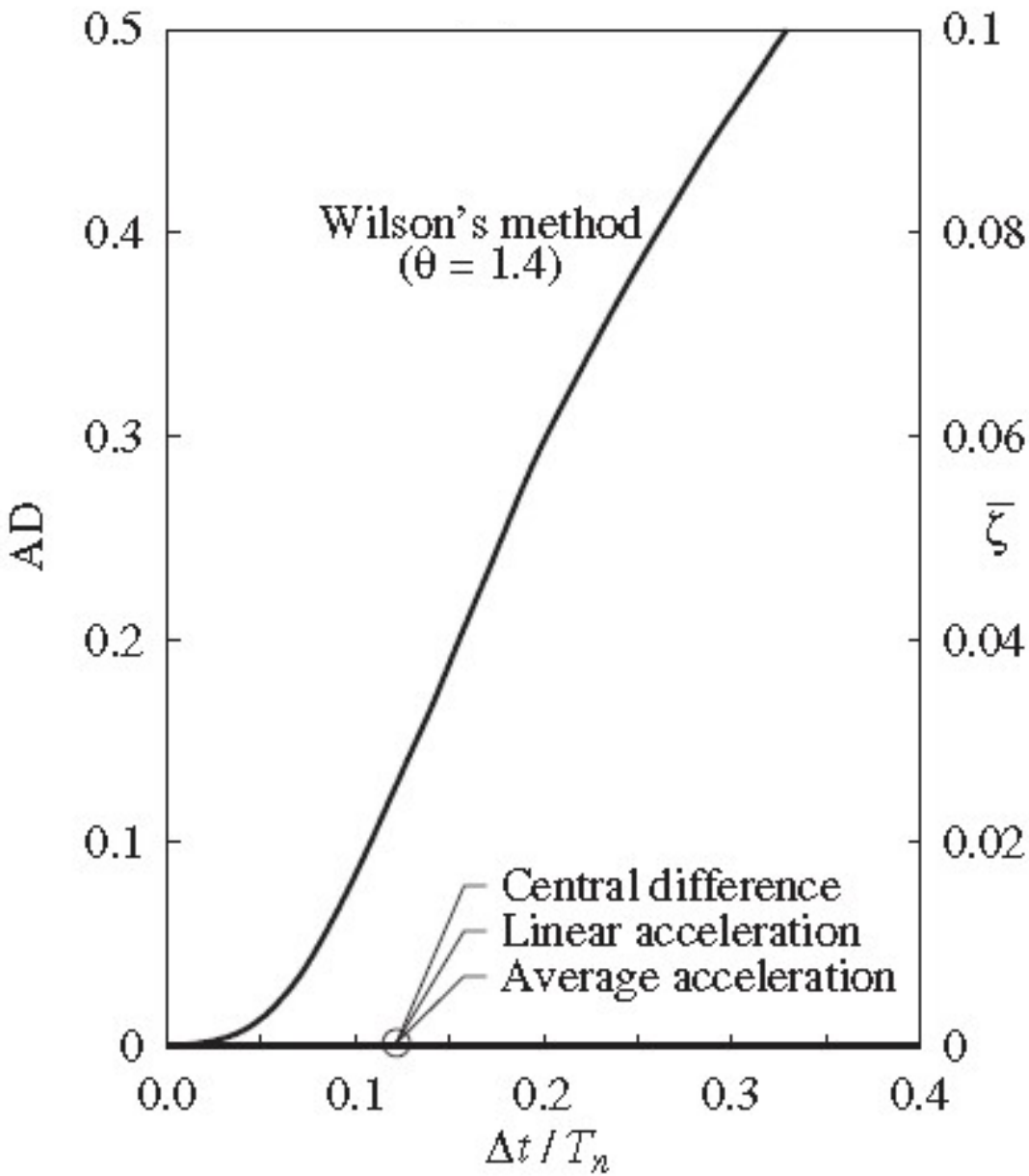
Free vibration solution for an undamped SDOF system by four numerical methods $\frac{\Delta t}{T} = 0.1$ and theoretical solution

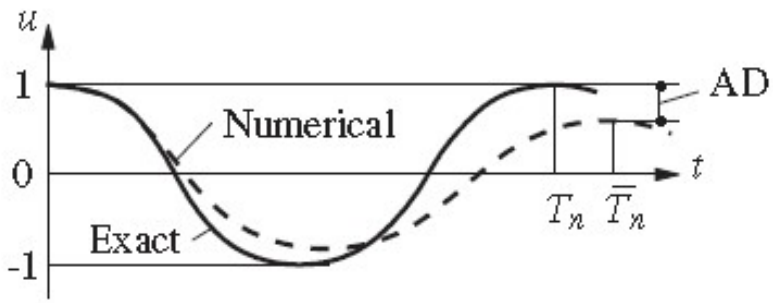


This comparison shows that some numerical methods may predict that the displacement amplitude decays with time, although the system is undamped, and that the natural period is elongated or shortened.
The next slides show the amplitude decay (AD) and period elongation (PE) in the four numerical methods as a function of the ratio $\Delta t/T$.

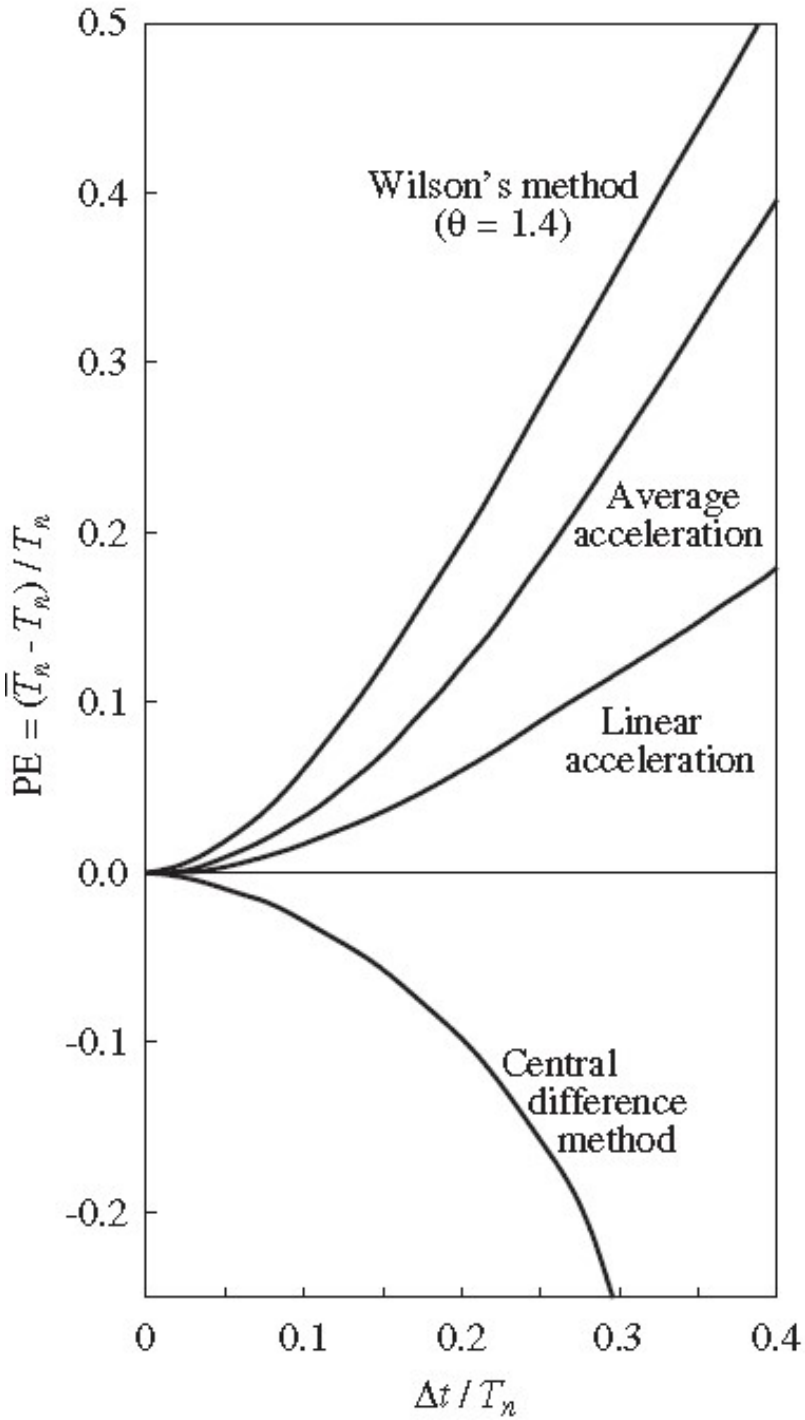
Note that the time step should also be short enough to keep the distorsion of the excitation function to a minimum. A very fine time step is necessary to describe numerically the highly irregular earthquake ground acceleration recorded during earthquakes; typically $\Delta t = 0.02\ s$ and the time step chosen for computing structural response should not be longer.

Amplitude decay (AD) versus $\frac{\Delta t}{T}$: numerical damping

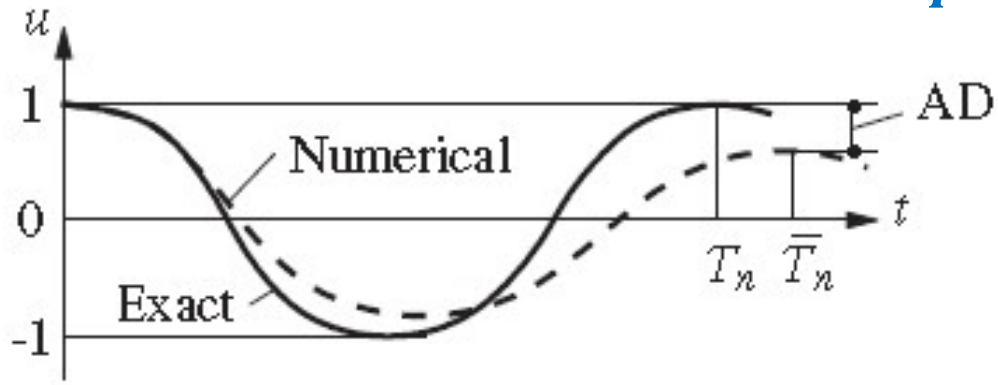




Three of the methods predict no decay of displacement amplitude. Wilson's method contains decay of amplitude, implying that this method introduces numerical damping in the system



Period elongation (PE)versus $\frac{\Delta t}{T}$



-Observe the rapid increase in the period error in the central difference method near the stability limit $\frac{\Delta t}{T} = \frac{1}{\pi}$. In this sense it is the least accurate of the methods considered.

- For $\frac{\Delta t}{T}$ less than its stability limit, the linear acceleration method gives the least period elongation. This property, combined with no amplitude decay, makes this method the most suitable one for SDOF systems.

One useful, altough unsophisticated technique for selecting the time step is to solve the problem with a time step that seems reasonable, then repeat the solution with a slightly smaller time step and compare the results, continuing the process until two successive solutions are close enough.

The present discussion regarding stability and accuracy of numerical methods applies strictly to linear systems.

Time stepping methods for MDOF systems are not reported in the present notes. Interested readers may refer to Chapter 15 of the Chopra book.